

House Building in Texas: a Geospatial Analysis of Climate Risk

MPlan City Planning Dissertation

Christine Langston

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Abstract

This dissertation focuses on the climate risk of new housing and real estate development. As climate induced natural disasters become more frequent and US population growth concentrates in several vulnerable urban centres, it is pertinent to understand the phenomenon and the government policies managing house building.

This research focuses on a case study of Texas to understand the intensity of population growth and housing development at the regional and local scale. Texas has four of the fastest growing urban areas and faces nearly half of the natural disasters that occur in the US. Previous findings establish that real estate across the US is overvalued, failing to incorporate climate risk accurately, and that house building is rapidly occurring in less regulated areas.

Government published datasets provide housing unit production counts, population counts, and estimated climate risk. The US Census Bureau publishes demographic data and the US Federal Emergency Management Agency (FEMA) published a National Risk Index (NRI) used in this study. Through quantitative and geospatial analysis, clusters of house building, population growth, and climate risk are found across Texas. At the local level, city zoning policies and house production are compared to climate risk to understand spatial distributions.

Analysis shows that Texas has grown by about one third in population and housing stock since 2000. Calculations discover that Dallas and Houston have high proportions of new housing units built in high-risk areas. In Dallas, zoning policies reserve land for residential development that skews towards low-risk areas.

With these findings, policy makers, homeowners, and prospective residents can understand the climate risk of new house building in Texas. Recommendations for policy include disincentivizing the building of housing and infrastructure in these high-risk zones to avoid increasing the amount of people and real estate investments exposed to risk from natural disasters.

List of Abbreviations

ACS - American Communities Survey

EPA - Environmental Protection Agency

FEMA - Federal Emergency Management Agency

NFIP - National Flood Insurance Program

NRI - National Risk Index

US - United States

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Unless noted, all tables and figures were generated by the author. The code for this analysis can be found in this public [Github Repository](#).

Chapter 1: Introduction

This dissertation studies the intersection of the climate crisis, housing supply, and land use policies in the United States. As more climate induced natural disasters occur, climate risk becomes important for all stakeholders to understand, including homeowners, renters, lenders and all levels of government. Understanding and assessing climate risk is the first step to preventing physical and financial damage and ensuring safe housing. Planners and communities need to understand the basic landscape of climate risk and how it currently interacts with real estate development and planning policies.

This research will explore the phenomenon of house building and climate risk in growing cities to develop findings that can provide knowledge to urban planners and development stakeholders. This research will answer the primary question **‘To what extent are local planning policies restricting new housing to high climate risk areas?’** by analysing house building within the scope of Texas as a case study.

Background

Each year, natural disasters destroy thousands of homes and cause billions of dollars of damage in the United States (National Centers for Environmental Information, 2024). This impacts personal safety, personal financial decisions, local economies, and financial decisions for industry investors and lenders, and the federal government. Specifically this past year, fires and back to back hurricanes have created a national conversation about climate change and housing in the United States. Newspapers have featured articles, including opinion pieces by researchers, concerned with the lack of public understanding of climate risk and its impact on housing (Keys, 2024; Lustgarten, 2024; Sastry and Sen, 2024). This has elevated the concern of residents and policymakers in the US to consider the supply of housing and the damage of natural disasters more strategically.

Climate Risk

As the climate crisis progresses across the world, some geographic areas are more at risk for climate disasters and extreme weather induced by climate change. The likelihood of events and the predicted damage of events such as storms, sea level rise, and drought are not distributed evenly across space. Nor is the world’s population evenly distributed across regions.

Scientists have put together advanced climate models that can predict where the most climate risk exists and the most damage will be caused. These include flood plain maps, and forecasting models of extreme temperatures, hurricanes, and storms that are both published by the federal government and private climate modelling companies (*Zillow*, 2024; *Zuzak et al.*, 2025).

Climate Planning

As more is known about climate change, governments and communities are making climate plans and policies for climate resilience. Climate planning is a term for ‘planning activities that seek to mitigate or adapt to climate change’ (*Lieberknecht*, 2022, p. 98), and can include any type of spatial or economic planning for mitigation or adaptation. Mitigation efforts include the reduction of greenhouse gases and their contribution to climate change. For example, housing and transport sectors are large emitters in the US (*Environmental Protection Agency*, 2022). Adaptation efforts can reduce vulnerability, for example, by preparing for or recovering from a climate event. For both mitigation and adaptation strategies, the dispersion of resources and infrastructure determined by the planning system is crucial.

Unfortunately, climate change endangers peoples’ homes and forces people to resettle or move when homes are destroyed. There is no doubt that communities are going through existential questions about their future.

Responsibility of Housing Impacts

There are huge emotional and existential risks to the people who are living in these high climate risk areas and climate events also pose a physical danger to housing and real estate assets, which can cause financial wreckage to the inhabitants and owners. Given that the disasters and damage happen unevenly across space, the question of how to manage and distribute the impact arises.

From a single climate event, there are huge expected losses of infrastructure, including housing. Climate risk has been found to be priced incorrectly into house prices, leading to overvaluation where the housing market participants do not have information on the true climate risk (*Gourevitch et al.*, 2023). The incorporation of climate risk into property values has already led to property value decline in risky areas and the creation of the term ‘climate gentrification’ (*Taylor and Aalbers*, 2022). Specifically for investors and lenders, physical risk from climate events is increasingly important to analyse for home insurance and residential mortgage credit

risk (First Street, 2025). In turn, the increased risk drives up insurance prices and the likelihood of needing to repair damage, driving up the cost of housing.

In the US, the Federal Emergency Management Agency (FEMA) insures over five million houses through the National Flood Insurance Program (NFIP), and has over \$20 billion of debt from disaster relief (Egger, 2023). While insurance for other disasters can be purchased through the private or state markets, the federal government offers flood insurance in areas where the private market does not. This flooding insurance through the federal government is paid for by taxpayers, so the federal government has the incentive to minimize the damage from floods because they are paying for it through insurance claims out of their budget (Egger, 2023). The existence of the NFIP has inspired rigorous research about flooding and housing in flood zones (First Street Foundation, 2021; Kousky et al., 2018; Sastry, 2022).

Growing Cities in the Sunbelt Region

The United States has recently experienced mass internal migration from other regions to the ‘Sunbelt’ region, which comprises the southern states from Arizona to Florida including Texas. This data has been observed for several years and is most evident in Census data from 2022 (Clark, 2023). This migration is partially explained by the allure of a lower cost of housing and economic opportunity in the southern US (Clark, 2023; Glaeser, 2011). However, it is difficult to accurately measure and distil migration down to a sole reason, much less generalise to conclude as the sentiment for all migrants.

The large movement of people into that region has resulted in the urbanisation and growth of urban centres, such as Dallas, Texas. The cities of the Sunbelt are growing both in population and in developed land area, with growth outward from the city centre characterised by urban sprawl. As cities grow, some states and local governments are taking planning action, such as permitting development for housing to provide for the new population and adjusting existing planning laws (Fechter, 2025). The US planning system revolves around land use designation in the form of ‘zoning’. Across the United States, local governments divide their jurisdictions into small ‘zones’ and designate the land use, building use class, and often building form by law.

If the land market uniformly priced in climate risk, lower risk land would be priced higher, therefore being advantageous to wealthier individuals. However, housing and land allocation is not purely market driven. Housing production is an economic development tool that local governments in the US usually have tight control over. The US planning system allocates land

development rights through zoning codes, which impacts land value. Zoning dictates housing form and density, driving an important distinction between single family homes and multi-family homes.

Social Justice

US cities already have spatial and temporal injustices. The US planning system has roots of being a system of discrimination and cause of spatial segregation and inequality in its cities. Not always, but zoning as a practice has historically created a form of segregation, as ‘Exclusionary Zoning’, and can create or reinforce geographical lines of socioeconomic and racial differences. For example, by restricting an area to only single-family homes and no flat blocks, it usually restricts that area to wealthier residents. This is unjust if economically less privileged people are limited to or only afforded housing in high-risk climate areas.

There also exists the intersection of climate justice with social justice. The concept of distributive justice acknowledges that climate impacts and vulnerabilities are unevenly distributed across a city and therefore across property owners and renters (Klinsky and Mavrogianni, 2020). Combining these climate risk and spatial inequalities for residents, there are many additional costs and decision factors associated with where someone lives.

This research will dive into understanding how local planning policies impact the distribution of new housing development across high climate risk areas. This dissertation raises the policy and planning question, ‘What can federal, regional, and local governments do to reduce the risk of house building and its residents in high-risk areas?’ This question is complicated by various political economy approaches that favour market to government intervention and exacerbated by urban areas experiencing growing populations.

Aims

The primary aim of this research is to understand if the existing housing market and housing regulation results in new housing units built in areas of high climate risk at a regional and local level. This comes from concerns about how local and regional governments are addressing climate change. Assuming that governments want to reduce the climate vulnerability of their residents and risk of public investments, local and regional governments could use this analysis to understand how climate disasters threaten their jurisdiction and reconsider land use. By

understanding trends of housing production and climate risk, planners could be better equipped to think about housing growth and development management.

A secondary aim of this research is to create a repeatable methodology with public, government-issued datasets that can be conducted across the United States. The research could then benefit policy makers, planners, and residents from other regions and be a guide for additional analysis.

Methodology Summary

This research uses a regional and city level case study to analyse housing production and climate risk. The methodology of this research could be applied to any geography in the United States, however, as justified later, Texas provides an interesting case study. This singular case study approach is limited because it does not allow for comparison and cannot be generalized to apply to the whole country. However, there is comparison at the city level, and this provides a reference point for population and housing growth.

This research uses quantitative analysis and secondary data sources to understand the spatial relationships between climate risk and house building. The author uses house building as a proxy for migration to the area and demand for housing from new residents. Mapping the data and looking at the factors spatially will help identify where there are clustered hotspots of population growth and climate risk. This can help local authorities and planners understand where the housing market and housing policy has shaped recent housing production.

Scope

For the scope of this dissertation, this research will focus on one region within the US. These research questions and themes could be studied anywhere in the world because, unfortunately, climate change is increasing natural disasters and increasing physical harm to housing across the world. This research focuses on recent history and extends back to 2000 to understand contemporary trends on a timescale common in the planning profession.

Research Questions

This dissertation examines the research question, ‘**To what extent are local planning policies restricting new housing to high climate risk areas?**’ through three objectives and by taking Texas as a case study:

- What is the status of housing and population growth across Texas?
- How do areas of high climate risk correlate with housing growth?
 - Where has new housing been built in Texas over the decade 2010-2020?
 - How is climate risk distributed across Texas?
- How do housing growth and local planning policies compare to climate risk at the local level?
 - Do land use laws restrict house building to high climate risk areas?

The following chapters will explore relevant literature and answer these research questions. Chapter 2 discusses the context and previous research that form the basis of this dissertation and Chapter 3 will outline the methodology, which includes the research design and methods. Chapter 4 will present the case study of Texas, with background on state land use policies, climate risk, and housing, and then Chapter 5 reviews the results and discussion of the quantitative and mapping data analysis. Chapter 6 presents the conclusion of the findings and extrapolates to relevant policy recommendations.

Chapter 2: Context and Literature Review

This literature review will cover contemporary research on four main topics: housing supply and policies, population change, housing costs, and housing and climate change, all in the geographical context of the United States. These topics are the areas in which this research contributes to further knowledge. Previous researchers have studied housing supply in the US (Glaeser, 2011; Hsieh and Moretti, 2019; Joint Center for Housing Studies, 2024) and the financial implications of climate risk on housing value (Gourevitch et. al, 2023; Hino and Burke, 2021; Sastry, 2022). However, the intersection of climate risk and housing production is understudied. This is crucial to understand the spatial distribution of new housing and if residents have climate safe housing.

Housing Supply and Policies

There is a large housing supply gap in the United States, according to The State of the Nation's Housing (Joint Center for Housing Studies, 2024), an annual report analysing the housing market in the US. Their findings come from aggregating, graphing, and mapping data from the US Census Bureau, real estate associations, and other third-party data providers.

The number of households in the US are growing. From 2022 to 2023, the number of households increased by 1.7 million, up to 30.3 million households. This increase is more than the average annual growth of the previous decade. Between 2010-2019, the annual average growth of households was 1.1 million. Approximately two thirds, or 85.9 million, of these households are homeowners, and the rest are renters (Joint Center for Housing Studies, 2024).

Lack of housing affordability and shortages of available homes have pushed state and local governments to incentivize housing supply growth and limit policy that inhibits house building. Cities are revising planning policies to allow the denser building of homes, away from single family homes, removing parking minimums, and easing the permit and environmental approval processes (Joint Center for Housing Studies, 2024). The report frames the zoning system as limiting housing stock because it restricts density and multi-family homes. One of its recommendations is to change zoning codes so that the development process is quicker and less risky, which would reduce the overall cost of development.

In the US, most land use policies are controlled at the municipal level and are localised to city limits (Wilson, 2025). Hsieh and Moretti (2019) explain that 'Housing regulations have

historically been perceived as having only local effects—affecting the quality of life and housing prices in the local community— and for this reason they have always been set at the municipality level’ (p. 32). They argue that society thinks of urban planning as a local issue, however, their findings indicate that local policies can have a larger impact on cost of living and economic opportunity and should therefore be a regional or national concern.

Economists have studied the implications of limited housing supply and effect of housing regulation and planning policies on macroeconomic trends and the spatial distribution of economic opportunities. Hsieh and Moretti (2019) found that housing market regulation by U.S. cities through land use and zoning laws impact the resource allocation of labour and where people live. They used quantitative data analysis to compare housing supply across different regions in the US and found that housing supply is limiting economic activity. They showed that increased housing stock could improve the economic output of a region. Between this benefit and increasing the affordability of an area, there is a clear motive for local and regional governments to increase house building.

Local policies impact the spatial distribution of new housing and areas with fewer policies have seen higher growth of new housing. In the US, states are broken down into counties which can include cities or towns. Unincorporated areas are sections of counties without established local governments, such as city or town. Researchers found that 42% of new housing was in unincorporated areas, although those areas only house 23% of the population (Bartik et al., 2024). This research pointed to housing market regulation in incorporated areas as being prohibitive. They found this by mapping housing unit density in four metropolitan regions in the US using 2022 ACS data. They use bivariate scatter plots to observe correlations and histograms of the variables to understand their distributions. They studied the spatial distribution of new housing developments and found them less likely to be within city limits (Bartik et al., 2024).

Population Change

Population changes across regions and cities can put additional strain on the housing supply and make government housing targets even harder to achieve. A major domestic migration trend in the United States shows the migration of people to a small number of cities in the southern states, according to a summary of the US Census data (Clark, 2023). Clark (2023) cites the low cost of living and economic opportunity as the motivating factor behind the

regional migration. A lower cost of living and opportunities for economic prosperity are two historic reasons for migration across the world and are the basis for humans pursuing a better quality of life.

Interestingly, data from the US Census shows that residential mobility, the rate at which people are moving around, whether between states or neighbourhoods, has been declining (Joint Center for Housing Studies, 2024). In greater detail, economists studied the migration of people to the Sunbelt over the last 50 years and they found that migration to the region has been decreasing over this time. They compared average temperatures to population statistics and found that hot, rural counties in the Sunbelt have been losing population, meaning people are moving to urban centres (Leduc and Wilson, 2024).

Changing populations create a challenge for local governments because demographics and tax bases change as well, and therefore the needs of the residents and the resources available are dynamic (Lustgarten, 2024). Recent research has uncovered ‘climate abandonment areas’, where people are leaving a place in favour of lower risk areas. This is already happening in Texas and Florida. About 3.2 million people have moved out of flood zones and about 7.5 million more are expected to migrate over the next 30 years (Lustgarten, 2024).

Regardless of the direction of population change, states in the US are still very populous and vulnerable to climate risk. The need to understand this phenomenon of population change and associated housing demand motivates the research conducted in this dissertation.

Housing Costs

Today, housing receives political attention because of its growing unaffordability. According to the State of the Nation Report, over 20 million home owning households, and 22.4 million renters are considered cost burdened in 2022 in the US (Joint Center for Housing Studies, 2024). That is over 23% of the home owning households. Housing burden or cost burden is defined in the US as when a household spends more than 30% of its income on housing and utilities, leaving less money for other necessities and disposable income.

This varies region to region across the US for a variety of economic factors. Glaeser (2011) explains that the cost of living difference between the coastal cities and the southern states is the reason for migration away from the prior and to the latter. Glaeser (2011) reasons that

deregulation and the relatively relaxed housing and planning regulation in southern states, such as Texas, keeps construction costs lower.

In addition to employment opportunities, amenities and resources, such as schools, food stores, banks, health centres are important in the quality of life and impact cost of life depending on the travel time and transportation cost. Sansone et al. (2023) researched the correlation of socio-economic factors and access to these crucial resources across census tracts in the US. They found that there is reduced access among communities of colour and poorer neighbourhoods.

Housing and Climate Change

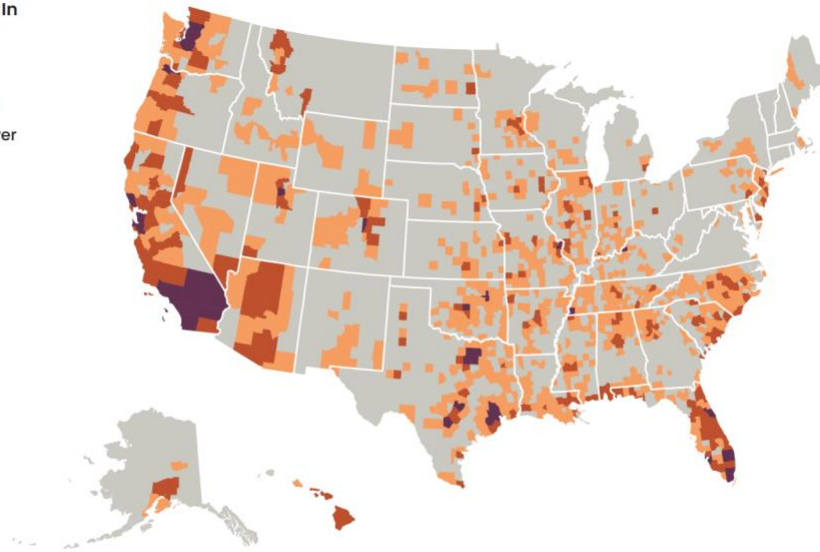
Climate change is one of the main uncertainties to predict and manage in the 21st century (Nijkamp et al., 2024) and it is making homeownership unpredictable and more expensive (Keys, 2024). The National Centers for Environmental Information tallies the number of Billion-Dollar weather and climate disasters and has counted 403 such disasters since 1980, including 27 in 2024. These disasters added up to more than \$2.915 trillion in costs and damages since 1980 (National Centers for Environmental Information, 2024).

In the United States, the Joint Center for Housing Studies (2024) found that 60.5 million housing units are subject to at least moderate climate risk, as shown in Figure 2.1. The report considers this a large number of households in danger. To understand the risk of housing for natural disasters, researchers from the Harvard University Joint Center for Housing Studies used FEMA's National Risk Index (NRI), which estimates a relative risk score based on estimated annual losses, social vulnerability, and resilience of a location.

More Than 60 Million Homes Are at Risk of Environmental Hazards

Number of Units In High-Risk Areas

- Under 10,000
- 10,000–49,999
- 50,000–199,999
- 200,000 and Over



Notes: High-risk areas are census tracts that have a relatively moderate, relatively high, or very high expected annual loss (EAL) score. EAL represents the average economic loss in dollars resulting from natural hazards each year. The number of units in high-risk areas is aggregated to the county level.
Source: JCHS tabulations of Federal Emergency Management Agency, July 2023 National Risk Index EAL data; US Census Bureau, 2021 American Community Survey 5-Year Estimates.

Figure 2.1: Map of United States colored by number of housing units at risk of environmental hazards (Source: Joint Center Housing Studies, 2024, p. 42).

Recently, from 2020 to 2023, insurance premiums have increased, showing risk for insurers is now estimated to be greater than it was before and cities have increased property taxes in order to pay for climate resiliency or repair infrastructure (Keys, 2024). There has been a large amount of research around the valuation of housing, losses, and risk for insurance due to climate change (Hino and Burke, 2021; Sastry, 2022; Gourevitch et. al, 2023). Flood risk is seen as the worst risk that causes the most deaths and damage across many geographical regions (Gourevitch et al., 2023).

A specific challenge is that climate risk can be difficult to price and housing market prices generally do not reflect true flood risk. There exists a wide overvaluation of real estate prices across the US (Hino and Burke, 2021; Gourevitch et al., 2023). Different jurisdictions in the US have different policies on disclosing flood risk during property sales and this impacts the discount rate that the risk has on the house price (Hino and Burke, 2021). Hino and Burke used secondary data sources and quantitative methods to answer their questions. They took properties and overlaid them with flood maps and census tracts and then panel regressions and difference in differences to understand causation. They used national datasets of flood plain

maps from FEMA, real estate data from national data aggregator, and census tracts and census block groups from US Census Bureau's TIGER data source.

Climate change impacts the property market by devaluing properties that are of high climate risk and increasing the value of properties that are safer. This can cause climate gentrification and drive poorer residents out of areas with lower climate risk in the future. Shi (2021) urges climate adaptation plans to consider the spatial distribution of different socio-economic groups in their proposals. Further than that, Shi points out that the distribution of wealth and land ownership is not considered in the climate reports, which 'pits frontline communities against middle-class, upper-class, small business, and public agencies in competing for scarce resources' (2021, p. 1468).

At a macro level, there is a lack of planning policy for climate change and climate risk (*The Economist*, 2024). Most funding currently goes to post disaster relief and recovery rather than the mitigation of climate change and limiting damage from natural disasters (Joint Center for Housing Studies, 2024). From a systemic level, the climate planning policy is therefore reactive rather than proactive.

The United States federal government offers flood insurance to protect properties, however they are not disincentivizing more development within floodplains (*The Economist*, 2024). This poses a specific threat for climate justice, as high-risk flooding areas are more likely to have low-income households (Egger, 2023). Low-income households are less able to pay for the mitigation and adaptation steps needed to protect themselves and their home from climate risk, including being able to afford insurance (Egger, 2023). Unlike other types of natural disasters, the US federal government offers flood insurance to homeowners under the National Flood Insurance Program (NFIP). The NFIP provides over 95% of policies for residential flood insurance, however only 30% of houses on floodplains have flood insurance (Kousky et al., 2018).

The Economist (2024) suggests that climate risk is not considered with enough weight when purchasing homes or deciding where to live. They say that homeowners are unaware of real climate risk, including when they purchased their property. There are many housing costs due to climate change taken on by the individual owner. Costs can be broken down into actions for mitigation and adaptation. For example, mitigation efforts are aimed to reduce the carbon emissions of the dwelling and therefore retrofitting to use less carbon intensive appliances or energy. For adaptation, property owners can invest in protection from climate events, for

example, flood walls or air conditioning. Unfortunately, paying for and recovering from climate event related damage is also a necessary cost of adaptation (*The Economist*, 2024). The Joint Center offers a similar solution, ‘An effective response requires both structural adaptations and financial resources to increase household, building, and land resiliency, and to reduce future risks by shrinking the residential sector’s carbon footprint’ (2024, p.7). However, they do not discuss or suggest shifting the location of housing development to less risky areas or any kind of climate retreat.

Not only are there injustices in how people are experiencing climate risk, but there is potential for future climate change mitigation and adaptation strategies to worsen inequalities. Climate change also poses a threat as a reason for migration and climate displacement. Shi (2021) presents evidence that ‘climate change presents a new driver of urban spatial restructuring’ (p. 1444). When considering real estate as an investment asset, climate risk can sway the potential benefits from owning to renting (Gourevitch et al., 2023).

Some US cities are using zoning policies to control housing development. For example, Norfolk, Virginia is trying to guide new development away from high-risk zones, which in this case is coastal flooding. They have seen more permits pulled in low risk zones, pointing at successful policy (Mendonca, 2023).

As this chapter explains, the policy factors of housing production, population change, and climate change have been well studied. However, no researcher has joined these topics together to intersect climate risk, growing population centres, and housing production. It is a novel research topic to study this phenomenon of housing production in growing cities and how it specifically exhibits in Texas. This dissertation also uses quantitative methods similar to the ones reviewed in the literature. Multiple of these sources that have researched the housing and of climate risk use secondary data sources and geospatial data analysis (Hino and Burke, 2021; Bartik et, al., 2024).

Chapter 3: Methodology

This chapter explains the methodology for this research by reviewing the research design choices taken and the methods used in the quantitative analysis. The secondary data sources are discussed and listed for reference. This dissertation research has been defined to study the spatial distributions of house building and climate risk. By choosing the specific case study region of Texas, the research examines the trends across the state and city levels. Data is analysed to understand the spatial distribution of population and housing units, if climate risk is evenly distributed across the state and cities, and to study if housing units have been built in high-risk areas.

A secondary aim of the research is to create a methodology of comparing variables that is transparent, relatively straightforward, and replicable in other geographies so that policy makers, planners, residents and prospective residents could benefit from this analysis.

Positionality

The author is from the United States and they were previously familiar with the planning system, zoning, and data sources, such as the US Census. This enabled the author to know about the phenomenon beforehand and which areas to research. This also allowed the author to find data sources comfortably, however also introduced bias and favouritism of data resources to the research. Although not from Texas, the author personally knows people in Texas affected by the weather events and climate disasters referenced in this dissertation. This sparked interest in the topic and region the author considers the impact of natural disasters with concern and sensitivity.

Research Design

These research questions are answered by looking back at recent history for trends in population and housing production and that compares to climate risk. This research is conducted through background research and geospatial data analysis techniques. The data analysis relies on statistical techniques used by economists and geographers and uses maps and bivariate graphs to visualize the trends.

Case Study Selection

For this dissertation, the research will focus on one state. This exact methodology could be repeated across the US in other regions with the same datasets because they are nationally available. A similar method could be derived for other countries to compare domestic migration, climate risk, and housing production.

Texas was chosen as the region of focus because of its rapid urbanisation and growth, and threat of climate disasters. Texas is in the central southern area of the US and is considered a member of the ‘Sunbelt states’. It has four of the ten quickest growing metropolitan areas, Dallas-Fort Worth, Houston, Austin, and San Antonio (Clark, 2023). It has a coastline and experiences hurricanes, extreme heat, and freezing weather (National Centers for Environmental Information, 2024). It also has national significance as the second largest by area and second most populous state. Additionally, the United States federal government and local governments within Texas have provided free and open data sets that are granular to small geographic units. This analysis is split between county and census tract spatial granularity.

Timeframe

The data is segmented by census data, which occurs every 10 years. This research focus on changes from 2000 to 2020. Some of this analysis focuses on population change from 2010 to 2020 to understand how quickly changes occurred over the most recent decade.

Secondary Data Sources

This research uses secondary data sources that are reputable and used by other researchers. This research takes advantage of existing high quality climate risk models and demographic data, so there is no primary data collection in this study.

Ethical Considerations

This research was conducted as desk research and secondary data analysis. The public data sources used are free and open to the public. There is no personal information in these datasets. Therefore, additional security precautions are not needed and the datasets have been stored locally on the researcher’s computer.

There is the possibility that this research into house and climate risk could uncover correlations or trends that would be undesirable for current homeowners and renters. In this case, the

findings might impact the real estate market and house pricing in the area of study. This is notable because real estate property is a large investment for many people and a negative effect on property value would negatively impact someone's financial stability.

Datasets

Data Quality and Availability

In the United States, there are public and freely available national datasets on climate risk and on demographics and housing provided by the federal government. The data is aggregated to various spatial areas and this study will focus on counties and census tracts. In order for data analysis to be interpretable and reproducible, the datasets used need to have widely available variables for all geographies and deemed of high quality (Nolte, et. al, 2021). For this reason, this research uses datasets that are commonly used by researchers in economics and geography. The list of datasets are in Table 3.1 below.

Population and Housing Data

There is a wealth of data available to the public and published by the government that is used for this analysis. The United States Census Bureau maintains and publishes data from the American Communities Survey (ACS) and the Decennial Census. The ACS is updated each year and provides aggregated socio-economic and housing market information. For this research, the population and housing unit counts will be used from the ACS and where not available, from the Decennial Census.

The Census Boundaries TIGER data provides digitized geography outlines that can be used for mapping the state, county, places, and census tract outlines. *Census tracts* are defined and updated depending on census population counts, redrawn every 10 years to capture the geography of 4,000 residents. Census tracts were last determined and updated in 2020 from the decennial census. In this case, *Places* are defined as a legally incorporated area or a designated population centre recognized by the state (Bureau of the Census, 1994). In this research, Place outlines are used to represent the boundaries of major cities for policy comparison. There are 1510 places in the state of Texas as of the 2020 Census.

Climate Risk Data

The Federal Emergency Management Agency (FEMA) publishes the National Risk Index (NRI) which provides risk of natural hazards for geographies across the US. This dataset is used by the Joint Center for Housing Studies (2024) and mentioned in the literature review above. The Center used it to estimate how many households are in moderate to high-risk areas, and for this reason, this dissertation uses the NRI for the same purpose.

The NRI dataset, last updated in March 2023, covers the entire United States and has values per county and census tract. The index includes an aggregate score and individual risk scores of 18 natural hazards: Avalanche, Coastal Flooding, Cold Wave, Drought, Earthquake, Hail, Heat Wave, Hurricane, Ice Storm, Landslide, Lightning, Riverine Flooding, Strong Wind, Tornado, Tsunami, Volcanic Activity, Wildfire and Winter Weather. This dataset has been put together to provide planners an understanding of the risk faced by their communities and then take action to mitigate those risks (Zuzak et al., 2025).

According to FEMA (2023), the NRI can help communities with the following goals: ‘identifying the need for more refined risk assessments’, educating ‘homeowners and renters’, ‘supporting adoption of enhanced codes and standards’ and forming disaster recovery plans (pg. 2). The audience of this dataset includes: ‘Public employees, local officials, community planners, academia, homeowners and renters, research institutions, property owners, real estate professionals, and insurance companies’ (FEMA, 2023, pg. 2). This would be an appropriate data source for planners to base housing and regulation decisions from and therefore a reason for being selected for this analysis.

FEMA is considered a credible data source for natural disaster risk because their datasets are heavily researched and used for analysing hazard risk (Zuzak et al., 2025). Other academic researchers on climate risk in the United States use FEMA flood maps (Gourevitch et al., 2023). The scope of this research does not include trying to estimate climate risk. There is lots of literature and research on how to best estimate climate risk using time series and probability models for each county in the US (Rasmussen et al., 2016).

The NRI has a risk score and a rating (1 to 5) indicating if an area is considered very low to very high risk, which can be seen in Figure 3.1 below. The risk is considered the product of likelihood and consequence of a natural hazard event. In this case, the creators of the NRI define likelihood as the Expected Annual Loss and the consequence as the Social Vulnerability divided by Community Resilience (Zuzak et al., 2025). The author of this dissertation would

like to note that this incorporates population numbers in the social vulnerability and the housing unit volume in expected annual loss, therefore making the NRI correlated with population and housing to some degree. This is a limitation of the score because the NRI is not completely independent of the variables studied. However, this score is being prioritised and used because of FEMA’s identification that it can be useful to planners for housing and regulation decisions and therefore it is relevant to the aim of this dissertation.

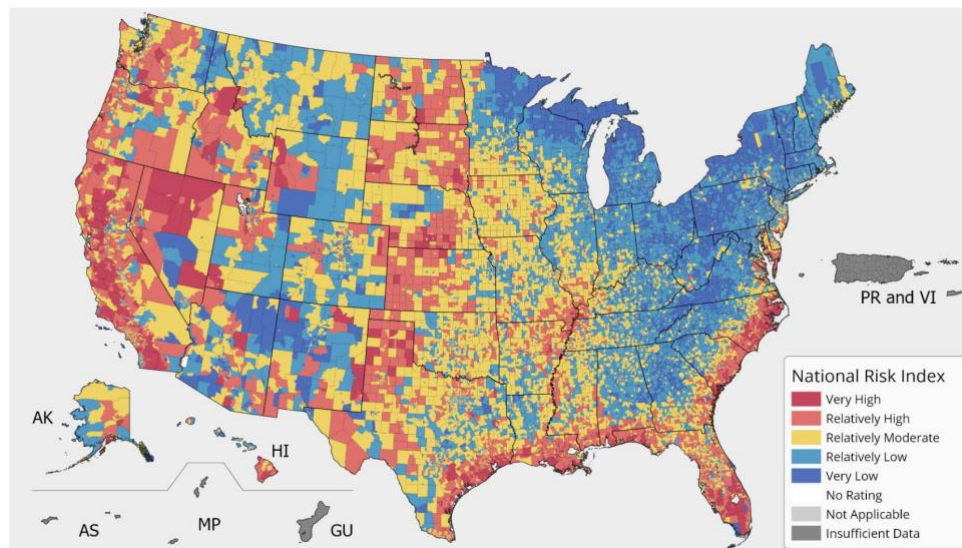


Figure 3.1: Map of the United States counties coloured by National Risk Index category

(Source: FEMA, 2025, Cover)

Zoning Code Data

On top of this, local governments publish zoning maps and use classes of properties as GIS files (City of Austin, 2022). In the last 10 years, the US has seen a wave of Open Data Portals sponsored by local governments, which publish and maintain GIS data for public use. See Table A-1 in Appendix for Dallas zoning districts considered residential and multi-unit residential.

Table 3.1 List of datasets used in research

Name of Dataset	Description	Use	Data source	Geographic Scale	Published or last updated date
ACS 5 Year	Demographics data for geographies in the US	Identify change in population and housing	US Census	Counties, places, census tracts	2020 ACS takes an average from data collected in 2015-2020
Decennial Census	Demographics data for geographies in the US	Identify change in population and housing	US Census	Counties, places, census tracts	2020
US Census TIGER Geography Boundaries	Outlines of jurisdictions that match the ACS data (counties, places, census tracts)	Determining spatial relationships between ACS variables	US Census	Counties, places, census tracts	2020
National Risk Index	Composite natural disaster risk based on likelihood of event happening	Identify relative climate risk	FEMA	Counties, census tracts	March 2023
Zoning Boundaries	Outlines of municipal zoning districts	Map and analyse residential zoning	City of Dallas	City	Accessed May 2025

Data Access

Census APIs were used to pull the necessary data from the US Census and ACS. This data resource required using an API key, which was free upon registration, to pull data by specific variable, geography, and year. The researcher then joined the variables together based on the geography identifier and then joined it to the polygon dataset representing geographic boundaries. Unfortunately, the variable names change from ACS to Census and over the years, so the variables used in this analysis are listed in Table 3.2.

All other datasets were available for download on their public website.

Table 3.2 Variables used from Census, ACS and FEMA datasets

Data Variable	Variable Name	Data Source
Housing Unit 2000, county and state	H001001	Census
Population 2000, county and state	P001001	Census
Housing Unit 2010, county and state	DP04_0001	ACS
Population 2010, county and state	DP05_0001	ACS
Housing Unit 2020, county and state	DP04_0027	ACS
Population 2020, county and state	DP05_0001	ACS
Population 2000, 2010, places	P001001	Census
Population 2020, places	P1_001N	Census
Housing Units 2000, 2010, places	H001001	Census
Housing Units 2020, places	H1_001N	Census
Housing Units built 2010-2019, census tract	B25034_003	ACS
Housing Units built 2020, census tract	B25034_002	ACS
Total Housing Units 2020, census tract	B25034_001	ACS
National Risk Index Composite Score, county and census tract	RISK SCORE	FEMA
National Risk Index Composite Rating, county and census tract	national_risk_index_rating_composite	FEMA
Building Value (\$), county and census tract	BUILDVALUE	FEMA
Population 2020, county and census tract	POPULATION	FEMA

Methods

The data manipulation and data analysis is performed in R, one of many tools that can be used for geospatial data analysis methods listed below (Bivand and Wong, 2018). The researcher chose R because of their familiarity with the programming language from previous work

experience and a UCL module. R includes the following libraries used in the analysis: sf for spatial data including polygons and geographic features, ggplot2 for graphing and mapping data visualisations, spdep for spatial dependence and autocorrelation.

All of the code used for this dissertation is publicly available from this [Github Repository](#).

Descriptive Statistics

The researcher used the housing unit counts, population counts, and fields from the NRI dataset to understand the spatial distribution of population, housing, and natural disaster risk. This research shows calculated differences, sums, and percentages to understand the changes over time and intensities of the phenomenon. This methodology is heavily inspired by the work of Bartik et al. (2024).

Spatial Autocorrelation

Spatial Autocorrelation is a popular method used to analyse the relationship of variables over space and is helpful for understanding how factors are clustered together spatially. This method is commonly used in geography, and environmental and social sciences (Goodchild, 2009; Maclachlan et al., 2024). It can be used to prove that a spatial trend exists, rather than variable values distributed randomly or uniformly across space. If a variable were randomly distributed across space, there would be no spatial dependence. However, many observed variables, such as climate risk, are spatially related, meaning that if a variable is high in one geography, its neighbouring geographies will likely have similarly high values as well (Goodchild, 2009). Spatial relationships are often easily spotted visually on maps, because similar colours representing the values will be clustered; however to understand spatial relationships in a deeper and more rigorous way, autocorrelation can be used as a quantitative and statistical technique to prove dependence (Gimond, 2024). Autocorrelation is ‘a fortunate characteristic of a wide range of spatially distributed phenomena’ (Goodchild, 2009, p. 414).

The method for calculating spatial autocorrelation starts with identifying the neighbouring geographies of the focus area and then calculating the mean of the neighbours and comparing that to the value of the focus area. Across the collection of geographies, the value of each geography and the mean value of its neighbours is then analysed. The correlation coefficient indicates. For example, if all the areas with a high value all had high means from their neighbours and vice versa with low values, there would be a strong correlation between a

variable's value and their neighbours' values, and evidenced as a dependency in values (Maclachlan et al., 2024).

There are multiple numerical calculation methods to calculate autocorrelation. These methods include the Moran's I and Getis Ord G^* statistics, which both range in information and interpretation. The first can indicate if clustering exists and the second can indicate if high values tend to cluster together (Bivand, and Wong, 2018; Maclachlan et al., 2024).

Moran's I is calculated in this analysis to understand the 'global' or overall clustering and determine if a value is spatially autocorrelated across the entire geography. A Moran's I statistical value of -1 indicates that values are dispersed, a value of 0 would indicate no pattern, and a value of 1 indicates clustering (Maclachlan et al., 2024).

Getis Ord G^* is calculated in this analysis to understand the 'local' clustering and hotspots. It helps indicate if a given area and its neighbours have higher or lower values than the overall or 'global' geography. Positive values of local Getis Ord G^* indicate high values are clustered together. So in the execution of this research, it helps identify counties and its neighbours that are clustered and higher valued or lower valued than the overall state of Texas (Bivand, and Wong, 2018).

Mapping

Data can be mapped to show the results of the statistical methods listed above and to show spatial trends, as demonstrated by Nolte et al. (2021) in their quantitative analysis of property sales and property prices and Hino and Burke (2021) in their study of flood maps and census tracts. Maps can be intuitive tools and easy to understand, however they also have their own set of limitations. Creating maps needs to be considered with well labelled data, legends, and the appropriate amount of information on each map (Maclachlan et al., 2024).

Synthesis

The descriptive statistics and maps can be used to compare total housing production and total population change for given areas over the time period of 2000-2020. Looking at the spatial distribution of those two variables over the state and two specific cities can investigate the phenomenon. The synthesis also includes looking at the distribution of the NRI rating across the state and city geographies. To pull the findings together, the distributions of existing housing are compared to distributions of growth of housing units across NRI risk areas. The

growth of housing is also considered across the city and in the context of zoning codes in Dallas and Houston.

Limitations

Only some analysis was able to be done at the county level and other analysis was at the census tract level because of the data available. The granularity of data and availability of data was limited. Not all datasets are available at census tract level. It's difficult to derive housing production by a change in housing units from year to year because the census tracts change every ten years with each census.

This methodology has been crafted to answer the research questions. The research design is oriented around finding spatial trends of house building and climate risk. The methods have been designed to be simple, effective, and use widely available datasets. This maximises the opportunity for this methodology to be repeated and applied to other geographical areas within the US.

Chapter 4: Case Study Background: Texas

Focusing on Texas can lead to understanding how housing in one region is changing over time and how housing growth correlates with the risk of climate driven natural disasters. To give context, the following sections explain recent trends of population growth, housing supply, natural disaster risk, and Texas specific planning and climate policy. Texas is a confluence of heightened climate risk, population gain, and new infrastructure construction, and therefore poses a unique case study.

Population Growth

The population of Texas is growing rapidly. Texas is the second largest state by population, behind California by about 8 million people (United States Census Bureau, 2023). Four of the top ten fastest growing US metropolitan areas are in Texas: Dallas-Fort Worth, Houston, Austin, and San Antonio (Clark, 2023). That has serious implications for its climate impact, as Texas emits the most greenhouse gases per state in the United States (Lieberknecht, 2022).

The Texas legislature has historically supported business and economic growth with incentives for businesses to expand in the state. In 2023, they passed an additional bill to give out tax abatements to companies that created jobs (Brooks Harper, 2023). Texas is seen as having an economic competitive advantage because of a reputation for lower rents and house prices (Fechter, 2025).

Housing Supply

Texas has seen a boom of house building to keep up with its population growth. Glaeser explains the expansion and growth of Texas by pointing out the low cost of living and housing. He writes, ‘The abundance of construction helps explain why Houston is so affordable’ (2011, p. 189). Between 2001-2008, 200,000 single family homes were approved in Houston, Texas (Glaeser, 2011).

Texas is known for building more homes than other states, a reputation comes from necessity and relatively light on regulations (Glaeser, 2011). Housing production can be seen in permit counts over time, as local planning departments give planning approval through permits in the US. According to the Figure 4.1 from the US Census Bureau (2022), Texas has issued more permits per year than any other state since 2006. In 2023, Texas authorized 232,373 permits, which is about 40,000 more than the second highest state (US Census Bureau, 2022). The rate

of job growth has increased three times since 2008 but construction of new homes has stayed the same (Fechter, 2024).

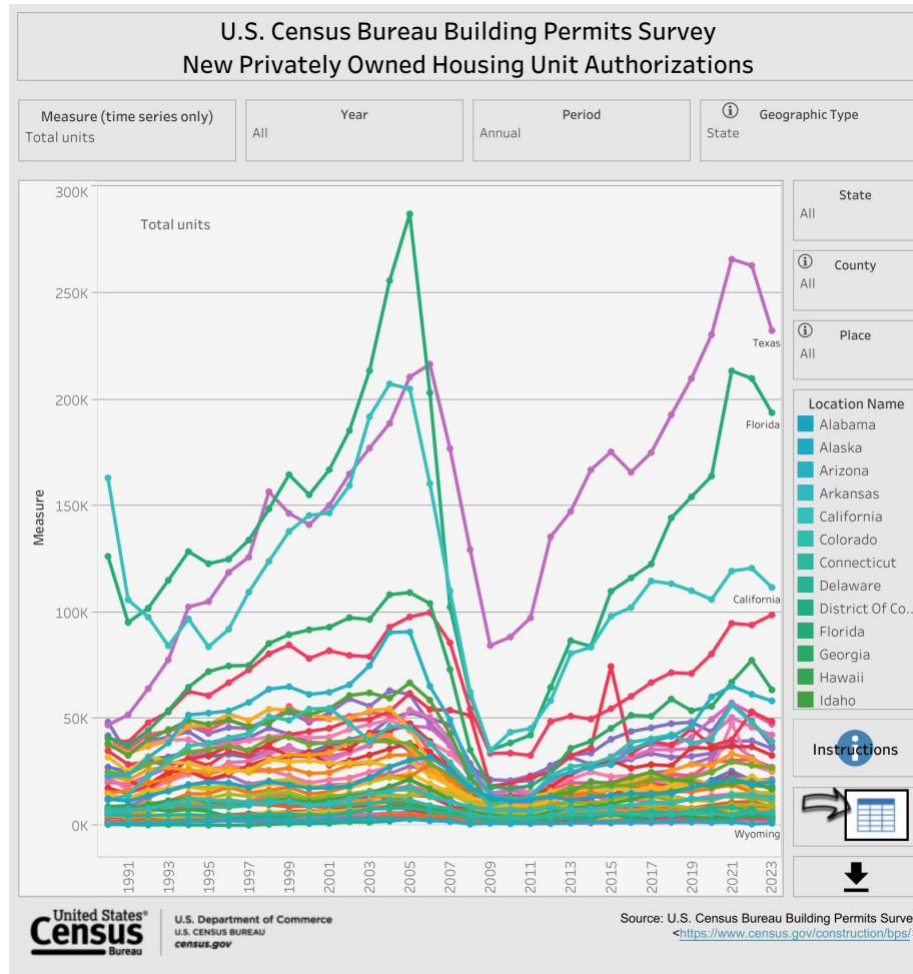


Figure 4.1: Graph of housing unit permits authorised by state from 1991 to 2023
(Source: US Census Bureau, 2022).

Natural Disaster Risk

The National Centers for Environmental Information keep track of extreme weather and climate disaster events that cause more than \$1 billion of damage. According to this list, Texas has suffered from a total of 190 events, made up of droughts, hurricanes (tropical cyclones), floods, wildfires, freezes, winter storms, and severe storms since 1980. This has resulted in an estimate of \$300-440 billion of damages, with about 57% of those costs attributed to hurricanes (National Centers for Environmental Information, 2024).

Overall, the United States has suffered from 403 weather and climate events over the same time period, so that means Texas has been affected by roughly half of those events (National Centers for Environmental Information, 2024). With increased population and housing development, planners and communities are confronted with the question of how to best protect themselves and infrastructure from this physical risk. This drives the motivation of this research to understand the distribution of climate risk over its population centres and to lay the foundation of knowledge to mitigate future risk through better infrastructure planning.

Planning Policies

The Texas state legislature is concerned about housing production as housing supply follows economic development as on the political agenda. Lawmakers have tried to restrict cities from development restrictions, however much of the power still lies in the hands of local politicians (Fletcher, 2025). Municipal governments in Texas have multiple levers to control, such as zoning for use and building type, minimum lot sizes, and parking minimums.

Zoning

Land use zones, which are codified by the municipal government, designate the use class type and size of buildings on land within the city boundaries. Residential, Commercial, and Industrial are basic use classes, but residential can also be broken down into single family homes, duplexes, apartments of 4 or fewer units, etc. Zoning prescribes the type of buildings and the residential density of neighbourhoods. When cities have zoning laws that require a significant portion of residential land be single family homes or low density homes, this restricts the quantity of units that can be built (Fechter, 2024).

Minimum Lot-Size

This type of planning policy limits the lot size and restricts the division of land between owners. The typical range for major Texas cities is between 5000 and 7500 square feet (Fechter, 2024), however many cities have been reforming this to increase housing density.

Parking Minimums

Development plans require surface parking allowances in proportion to the number of residents. This is a planning decision to allow for enough space for cars. However, given the space that parking can take up, these policies are in conflict with building additional units and maximizing the residents and density of a given plot of land.

Breakdown By City

Houston does not have zoning codes and allows all development uses across the city (Fechter, 2024). In 1998, they reduced their minimum lot size from 5000 square feet to 1400 square feet in some urban areas, and 3500 in most of it. They expanded the 3500 minimum lot size to cover the entire city in 2013. This ended up impacting the development of units because the city built 25,000+ units between 1999 and 2016 on lots smaller than 5,000 (Minott, 2023).

Dallas has a typical Zoning code with residential districts and specific districts that allow multi-family units (See Appendix Table A-1). Of the land that Dallas has designated for residential, 96% of that land is for single family homes (Fechter, 2024).

Climate Policies

Texans are found to believe in climate change and value the environment, however there is divergence between personal beliefs and state government action. Lieberknecht (2022) surveyed Texas residents about beliefs behind climate change and found, ‘Respondents’ beliefs starkly contrast with the level of attention climate planning receives at the state level’ (p. 106). From the survey, people agreed that climate change was causing negative impacts, including extreme weather events, and people valued the environment. From this study, the research found climate change beliefs can vary with space and from city to city within Texas, which they attribute to the co-location of people with similar backgrounds and ideological beliefs in the same geographic location (Lieberknecht, 2022).

At the state level, the government is doing little to mitigate climate change and is limiting local government’s ability to act. Texas is the state that emits the most greenhouse gasses, mostly coming from transportation and the electric power industry (Environmental Protection Agency, 2022). However, cities in Texas are joining nation-wide programs for climate resilient cities and creating climate plans (Lieberknecht, 2022). The state legislature took away autonomy from cities for making their own climate decisions and will not allow cities to pass climate charters unless the charter is approved by the state first. This prevented one city, El Paso, from setting goals to achieve 100% renewable energy (Douglas et al., 2023).

In 2023, Texas failed to pass statewide bills that were aimed to reduce greenhouse gas emissions and reduce energy usage, which would be steps of climate change mitigation. Although they did allocate \$2 billion towards water infrastructure and flood prevention

(Douglas, Foxhall, and Martinez, 2023). About \$1 billion will go towards improving water infrastructure that will help use water most efficiently, to help the region face droughts as a climate adaptation strategy. This will also make the water system more resilient to freezes and storms that have caused damages in the past. The other \$1 billion will be used to prevent flooding, including a large sea wall off the coast of the Houston metropolitan area and funding to implement the state flood plan (Douglas, 2023).

Texas passed its first state flood plan in 2024, which outlines the existing flood risk and recommends policy changes to reduce flood risk and plan for flood mitigation (Texas Water Development Board, 2024). Texas is relatively ahead of other states at the property level, because Texas policies require all three types of flood risk disclosure during the sale of real estate: flood plain, flood damage, and flood insurance, which is not required in all states (Hino and Burke, 2021). However, these types of disclosures are not necessarily made before offer or the price is set, so the climate risk is not always incorporated into the property sales price.

In conclusion, Texas is a region of intense population growth and intense permitting, facing extreme natural disasters with a diverse collection of local planning and climate policy approaches. This makes for a prime research case study. Focusing on Texas can help understand the interaction between a region with high housing demands, high housing production, and relatively little restrictions from climate policy.

Chapter 5: Results and Analysis

This chapter presents the findings from quantitative data analysis to answer the research question, ‘**To what extent are local planning policies restricting new housing to high climate risk areas?**’. This data analysis uses population, housing, and climate risk data from secondary data sources. It focuses on population and housing growth across Texas and in two of the largest cities to understand the intensity of housing development compared to climate risk, as represented by FEMA’s National Risk Index (NRI).

First, there is an overview of the state of Texas from 2000 to 2020 to understand population and housing production trends. The analysis covers the spatial distribution of the population change and housing unit change and presents results from autocorrelation to establish that clusters of high growth exist. Clusters of fast-growing counties are identified. Then, the distribution of NRI values across counties are analysed and clustering is found.

After that, in the City Level Analysis, the results present the fastest growing cities in Texas before centring on Dallas and Houston. The analysis time frame is shortened to 2010 to 2020 to focus on the most recent activity and the population, housing, and NRI values are considered at the census tract level. For each city, the spatial distributions of housing density and NRI are analysed. In Dallas, the zoning districts are compared to the NRI Rating of census tracts to understand the climate risk of recently built residential and multi-family units.

State Level Analysis

Population and Housing Change

Analysis of US Census state level data shows Texas grew in population and in housing units from 2000 to 2020. As listed below in Table 5.1, In the first ten years, the state grew by about 3.5 million people, which is a 16.6% ten-year growth rate; between 2010 and 2020, it grew by about 4.3 million people at a rate of 17.8%. Over the 20 years, the state grew by about 7.8 million people, at a rate of 37.3%. This means the population grew by a third in a generation, which is a huge change for any region or government.

Over these two decades, the housing stock grew just over 2.9 million housing units, or a 36.2% increase. According to the statistics in Table 5.1, housing stock grew faster in the 2000-2010 period than 2010-2020 and outpaced the population growth when comparing percentage

change. However, in the second decade, from 2010 to 2020, housing fell behind population growth. Over the 2000-2020 period, one unit was built for every 2.63 new inhabitants. The previous statewide average was 2.56 people per unit in 2000. These numbers are quite similar, so no dramatic shift can be concluded and this could be explained by several changing demographics, such as evolving average household sizes or fewer people living alone.

To further understand the growth of Texas population and housing, the population and housing can be studied spatially by county. The following analysis presents the spatial distribution of population and housing change.

Table 5.1 Texas state statistics

	2000	2010	2020	2000-10 Change	2000-10 Percent Change	2010-20 Change	2010-20 Percent Change	2000- 20Chan ge	2000-20 Percent Change
Population	20,851,820	24,311,891	28,635,442	3,460,071	16.6%	4,323,551	17.8%	7,783,602	37.3%
Housing Units	8,157,575	9,718,470	11,112,975	1,560,895	19.1%	1,394,505	14.3%	2,955,400	36.2%

(Data Source: US Census, 2020)

Population

From calculations, the mean population change from 2010 to 2020 across counties is 6.55%, and the standard deviation is 17.06, which means there is considerable variation in population change across counties. Given that the statewide population change over this period was 17.8%, according to Table 5.1 above, some highly populated counties would have had to grow above the mean of 6.55% to account for the overall growth.

Figure 5.1 maps of the percentage population change from 2010 to 2020 and shows slight regional differences in population change. There is one dark green county in the west, which shows a level of population change above 150%. From Figure 5.1, spatial differences can be seen but it cannot be determined if there is significant clustering of population change.

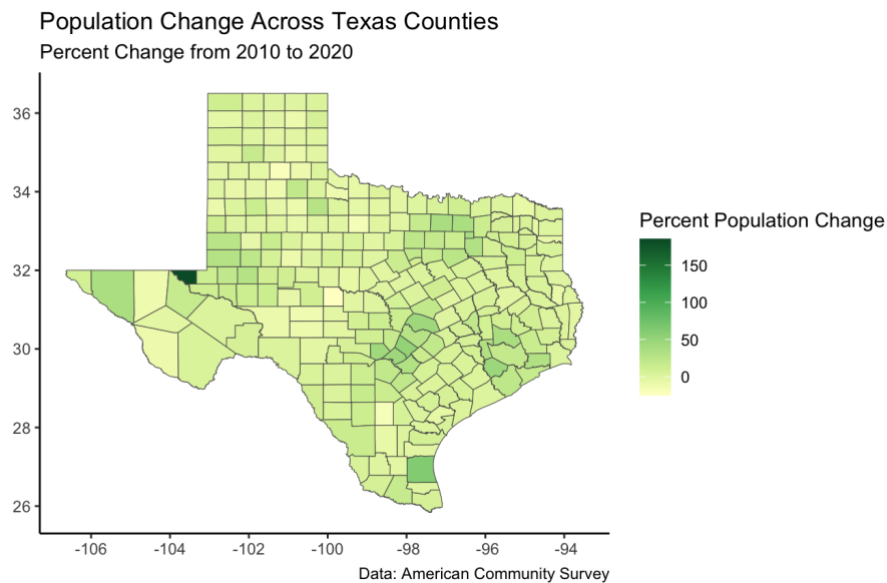


Figure 5.1: Map showing the percentage change of population per county from 2010 to 2020. (Data Source: ACS, 2020)

Given that Figures 5.1 did not show significant clustering of population change, the researcher performed a spatial autocorrelation test, which can conclude if there are statistically significant clusters and where ‘hot spots’ of growth exist. The autocorrelation compares each county to its local neighbours and determines if the variable value, in this case percent population growth, is similar to its neighbours.

Autocorrelation is considered to show clusters if the Global Moran’s I statistic is larger than 0 and the p-value is less than 0.05. In this case, Moran’s I was calculated to be 0.24745 with a p-value of $1.522e-14$, which means there is statistically significant clustering across Texas. The Local Getis Ord G^* statistic was calculated for population change for each county and is mapped below, in Figure 5.2. The map visualises several dense clusters of dark green counties: four primary clusters, with one in the western side of the state and three in the eastern side. There are indeed clusters of population change, discussed more later in the chapter, and this disproves that there is uniform change in population across the state from 2010-2020.

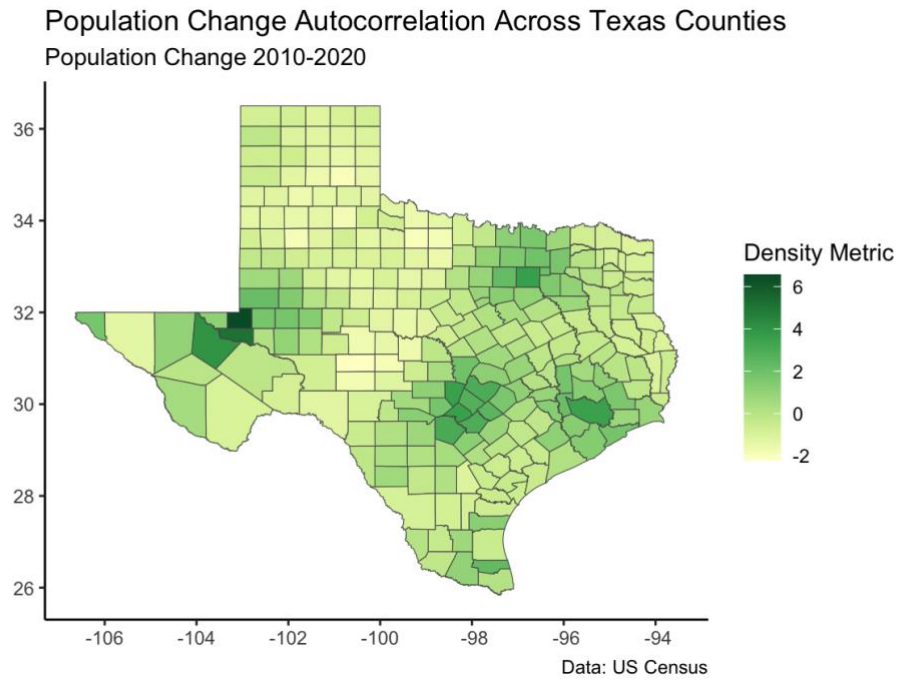


Figure 5.2: Map showing the autocorrelation values of population change per county from 2010 to 2020 (Data Source: US Census)

Housing

From calculations, the mean housing unit change from 2010 to 2020 across counties is 5.49%, and the standard deviation is 9.15, which means there is considerable variation in housing unit change across counties. Given that the statewide housing unit change over this time period was 14.3%, according to Table 5.1 above, some densely built counties grew faster than others to account for overall growth.

Figure 5.3 maps the percentage housing unit change from 2010 to 2020, and shows a range of colours with several dark green counties, signalling a very high level of housing unit change of about 40%. From Figure 5.4 spatial differences and clusters can be observed.

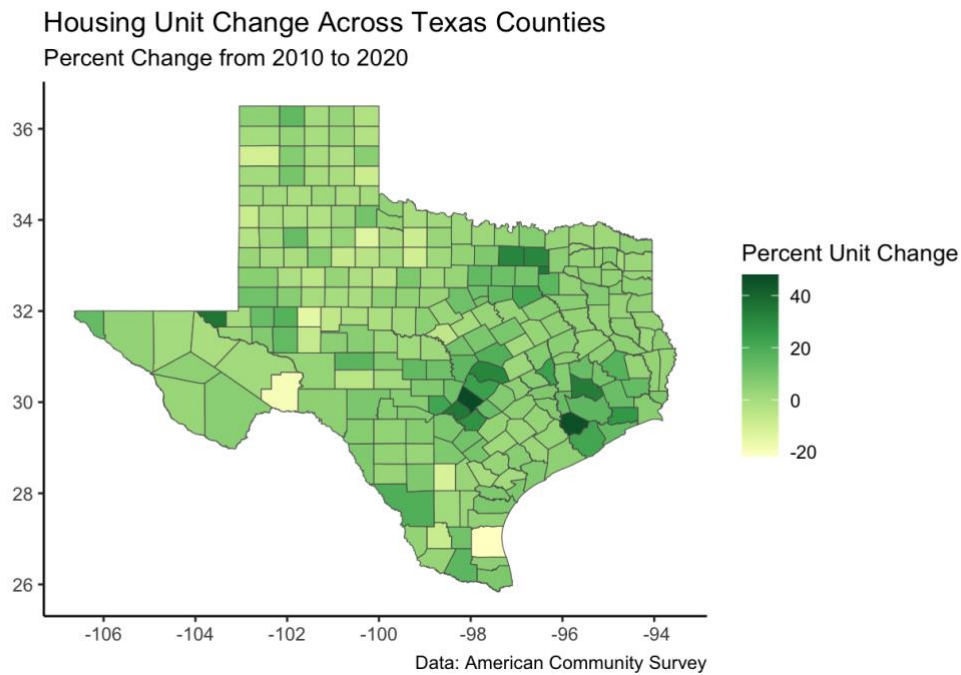


Figure 5.3: Map showing the percentage change of housing unit count per county from 2010 to 2020.
(Data Source: ACS, 2020)

To confirm the significance of clusters of housing unit growth and understand them more in depth, the autocorrelation tests were performed. For housing unit change, Global Moran's I was calculated to be 0.33326 with a p -value of $2.2e-16$, which means there is statistically significant clustering. The Local Getis Ord G^* statistic was calculated for housing unit change for each county and is mapped below, in Figure 5.4. The map shows several strong clusters around three areas in eastern Texas, centred around Dallas, Houston, and the San Antonio-Austin metropolitan regions. There is an additional, although weaker cluster in the western side of the state in the area that had high population growth. This disproves that there is uniform change in housing units across the state from 2010-2020. In general, the location of these clusters resemble the clusters shown in the population change autocorrelation in Figure 5.2. This makes sense because housing units and population tend to grow together and are interdependent.

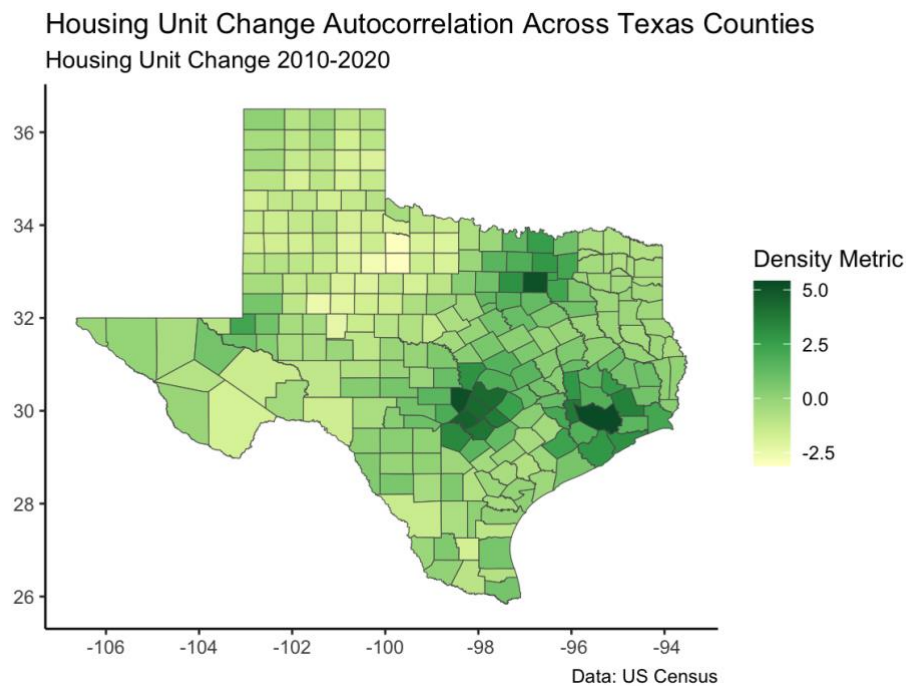


Figure 5.4: Map showing the autocorrelation values of housing unit change per county from 2010 to 2020. (Data Source: US Census, 2020)

Clusters of Growth in Texas

To understand the intensity and clustering of population and housing growth, the research investigates data at the county level. Counties are jurisdictional boundaries ranging in population values, so what matters over the absolute number is the relative change over time, which can be understood by looking at the percentage difference between years.

Below in Table 5.2 are the fastest growing counties by population from 2010 to 2020. Over this period, the fastest growing county by population was Loving County at a rate of 185%. This county is in the western side of the state and appeared dark green in the population change map above (Figure 5.1). However, the population of this county was less than 500 people so it will not be focused on in this analysis, which aims to understand urban growth. The second fastest growing county was Kenedy County, which also has less than 500 people. The next five

fastest growing counties were: Hays, Fort Bend, Williamson, Comal, and Kendall. These are all counties within the four fastest growing metropolitan regions according to Clark (2023):

- Hays and Williamson are both in the Austin metropolitan region
- Fort Bend is in the Houston metropolitan region
- Kendall and Comal is in the San Antonio metropolitan region

In Figure 5.5 below, the size of populations and growth rates become visually apparent. Fort Bend and Williamson are the largest counties of the seven and both have 2020 populations over 500,000 people. Below, Figure 5.6 shows the seven fastest growing counties outlined with colour overlaid on the map of the Texas population change.

Table 5.2. Fastest growing Texas counties, sorted by 10-year population change 2010-2020.

	Population					Housing Units		
Jurisdiction	2000	2010	2020	2010-20 Change	2000	2010	2020	2010-20 Change
Texas	20,851,820	24,311,891	28,635,442	17.78%	8,157,575	9,718,470	11,112,975	14.35%
Kendall County	23,743	31,515	45,491	44.35%	9,609	13,508	16,488	22.06%
Comal County	78,021	102,836	148,921	44.81%	32,718	45,099	60,661	34.51%
Williamson County	249,967	391,715	570,437	45.63%	90,325	151,524	199,061	31.37%
Fort Bend County	354,452	541,983	790,892	45.93%	115,991	178,626	260,936	46.08%
Hays County	97,489	146,439	222,827	52.16%	35,643	55,188	81,696	48.03%
Kenedy County	414	241	391	62.24%	281	282	221	-21.63%
Loving County	67	41	117	185.37%	70	64	87	35.94%

(Source: ACS)

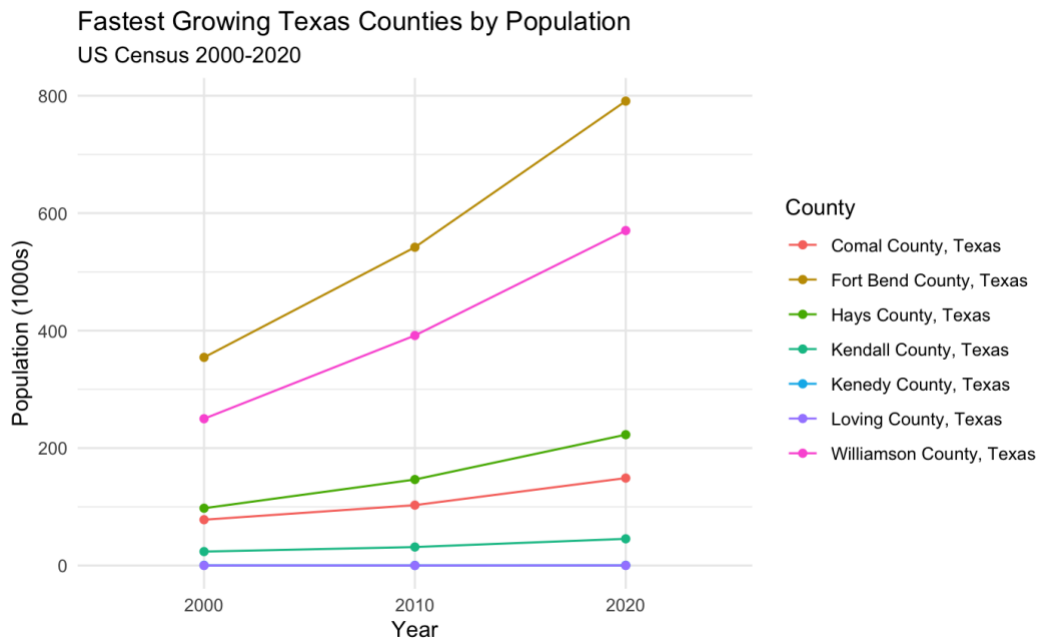


Figure 5.5: Fastest growing Texas counties by population from 2010-2020. (Data Source: US Census)

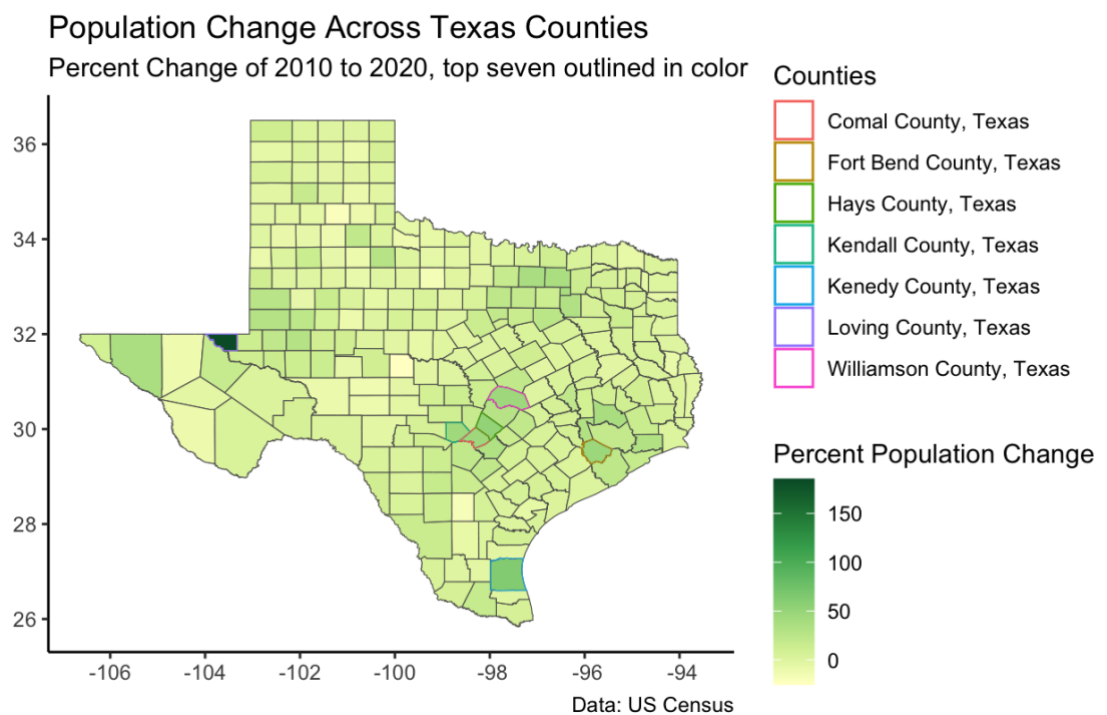


Figure 5.6: Map showing population change across Texas counties from 2010 to 2020, with the colours outlining seven counties of highest population change by percentage. (Data Source: US Census)

However, these fastest counties of population growth are not the fastest counties growing by housing units. In Figure 5.7, the graph shows the fastest growing counties by housing unit

change over 2010-2020, which includes Montgomery and Rockwall counties instead of Kenedy and Kendall. Looking back at Table 5.2, it is apparent that there are several counties where population growth by percent change outpaced housing growth. Kendall, Comal, Williamson, and Kenedy are all counties where the population growth outpaced housing unit growth by at least 10% between 2010-2020.

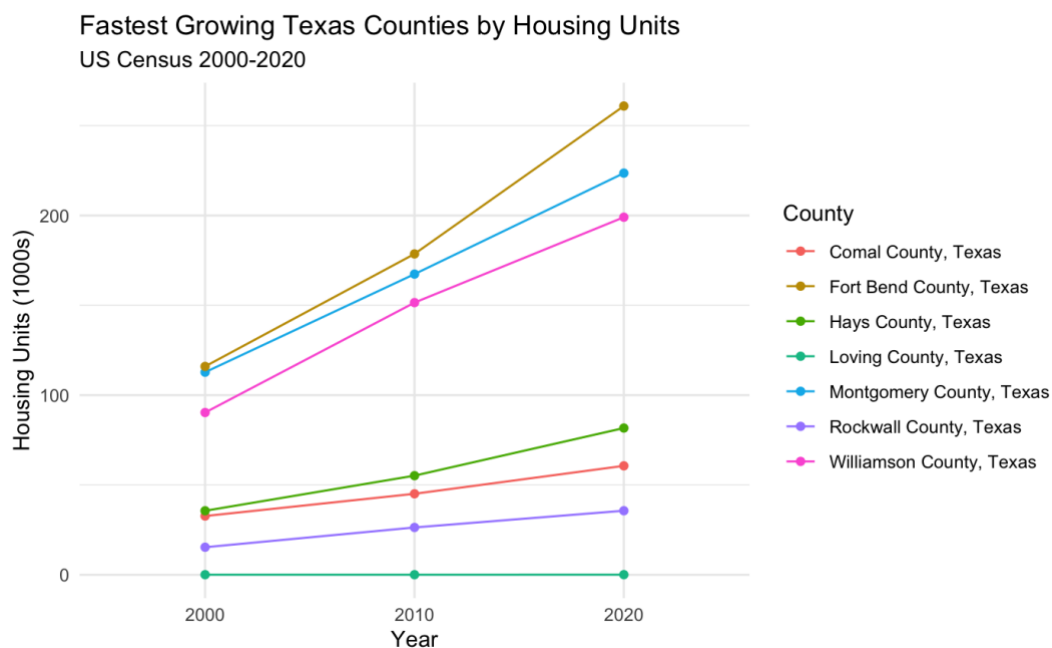


Figure 5.7: Graph showing housing unit count change across Texas counties from 2000-2020.
(Data Source: US Census)

Climate Risk

The National Risk Index is a composite risk score considering many types of natural risk hazards and is used to represent climate risk. It is found to vary across the state of Texas, which acknowledges regional differences in natural disasters and population and infrastructure, which are incorporated as estimated losses and social vulnerability. As seen in Figure 5.8, the eastern side of Texas is dark green and has high NRI scores. There are also higher scores along the coast, as marked by darker coloured counties. Five of the seven counties with highest population change have high-risk scores, as shown in Figure 5.9.

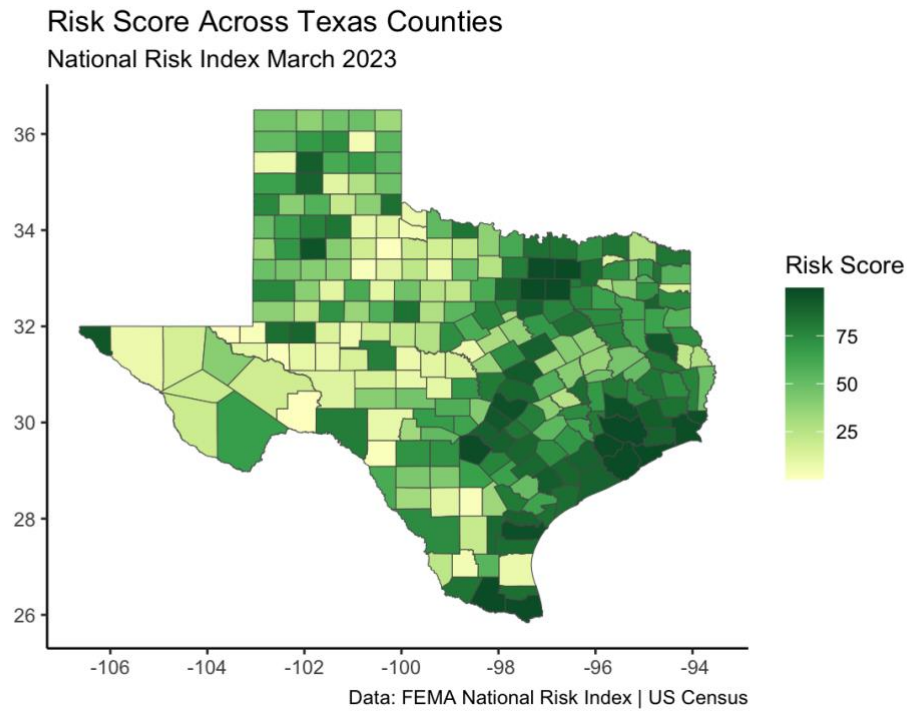


Figure 5.8: National Risk Index score mapped across Texas counties. (Data Source: FEMA; US Census)

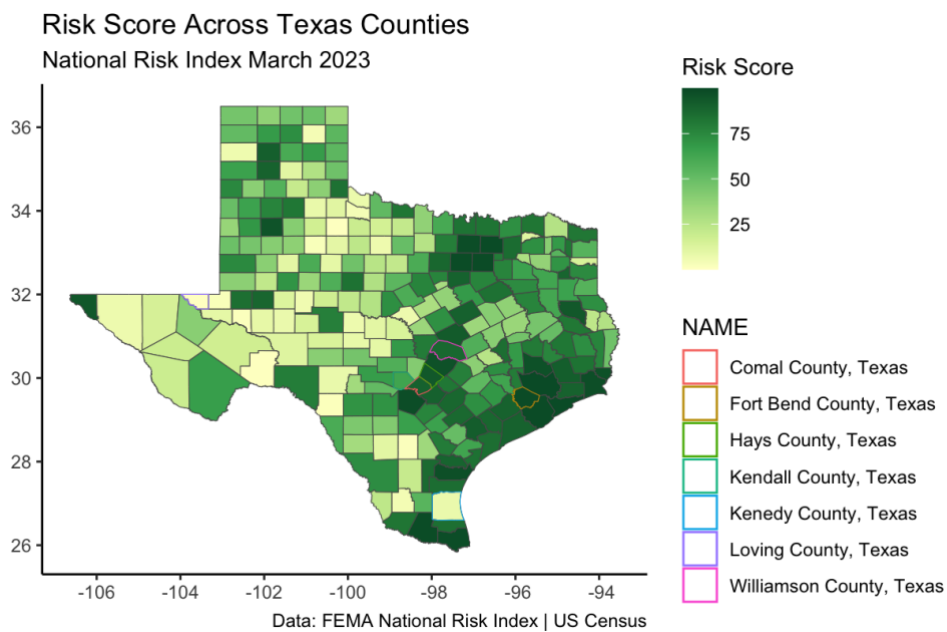


Figure 5.9: National Risk Index score mapped across Texas with the fast-growing counties outlined. (Data Source: FEMA, US Census)

Autocorrelation identifies the spatial clustering of NRI in Texas. The Moran's I Global statistic for county NRI risk score is 0.39504 with a p-value of $2.2e-16$. Because this Moran I statistic is greater than 0 and the p-value is less than 0.05, this means that there is statistically significant

clustering across Texas. The Local Getis Ord G^* statistic was calculated for each county and is mapped below, in Figure 5.10. The map shows several clusters across Texas and darker clusters in eastern Texas and the northwest. This disproves there is uniform climate risk across Texas, and in fact identifies higher risk around the main metropolitan regions and clusters of growth previously discussed.

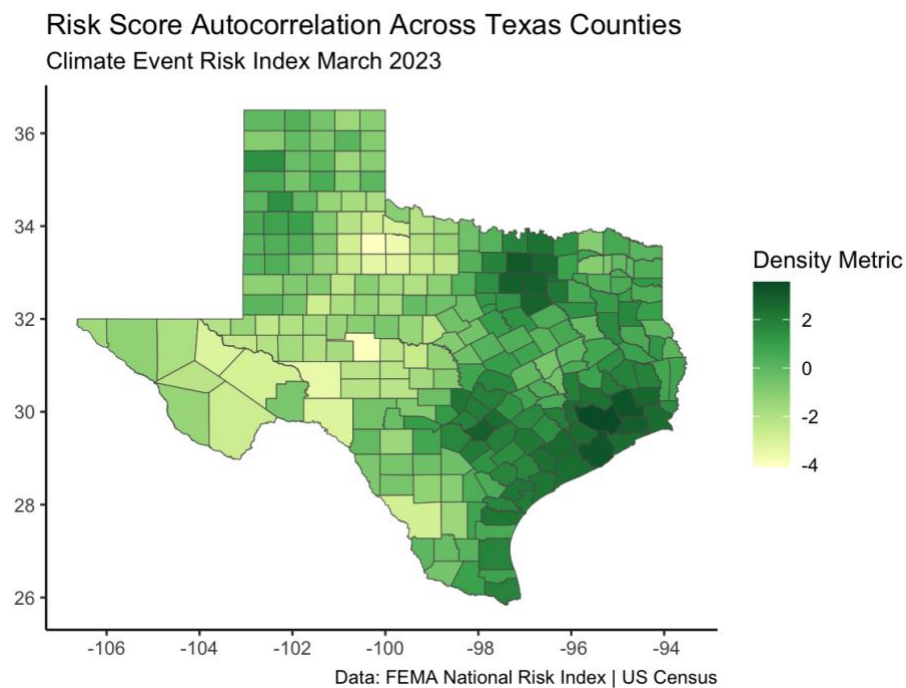


Figure 5.10: Map showing the autocorrelation values from NRI score across Texas counties.
(Data Source: FEMA; US Census)

City Level Analysis

In the US, local planning policies are set at the municipal level, and it is important to consider housing and population growth at that scale. This city level analysis will proceed to look at the fastest growing places and dive into three municipalities to compare the housing and population growth between them. This analysis will focus on the three most populous cities as of 2020, which also were chosen because they represent three distinct metropolitan areas and different planning policy landscapes.

Over the past twenty years, the top ten cities by population have shifted in order but have mostly comprised of the same ten cities. In 2000, the top population cities were: Houston, Dallas, San Antonio, Austin, El Paso, Fort Worth, Arlington, Corpus Christi, Plano, and Garland. In 2010, that order shifted to Houston, San Antonio, Dallas, Austin, Fort Worth, El Paso, Arlington, Corpus Christi, Plano, Lubbock. In 2020, the order stayed the same as 2010 except the 10th city, Lubbock, was replaced by Laredo (US Census).

Below in Table 5.3 are the ten fastest growing places, as sorted by absolute population change from 2010 to 2020. Houston is the fastest growing city by absolute population change from 2010 to 2020 and the largest city in Texas as of 2020 with 2.3 million people. The populations of Houston, Fort Worth, Dallas, Austin, and San Antonio make up 24% of the total state population in 2020. Four of the cities have grown rapidly since 2000, by more 100% percent: Frisco, McKinney, Pearland, and Conroe as shown in Table 5.3.

Table 5.3. Fastest growing Texas cities, sorted by absolute population change 2010-2020

Place (2022 population ranking)	Population 2000	Population 2010	Population 2020	Absolute Population Change 2010-2020	Percent Population Change 2010-2020	Absolute Population Change 2000-2020	Percent Population Change 2000-2020
Houston (1)	1,953,631	2,099,451	2,304,580	205,129	9.77%	350,949	17.96%
Fort Worth (5)	534,694	741,206	918,915	177,709	23.98%	384,221	71.86%
Austin (4)	656,562	790,390	961,855	171,465	21.70%	305,293	46.50%
San Antonio (2)	1,144,646	1,327,407	1,434,625	107,218	8.08%	289,979	25.33%
Dallas (3)	1,188,580	1,197,816	1,304,379	106,563	8.90%	115,799	9.74%
Frisco (14)	33,714	116,989	200,509	83,520	71.39%	166,795	494.74%
McKinney (17)	54,369	131,117	195,308	64,191	48.96%	140,939	259.23%
Irving (11)	191,615	216,290	256,684	40,394	18.68%	65,069	33.96%
Pearland (26)	37,640	91,252	125,828	34,576	37.89%	88,188	234.29%
Conroe (44)	36,811	56,207	89,956	33,749	60.04%	53,145	144.37%

(Data Source: US Census)

As seen in Figure 5.11, all the top ten fast growing places are geographically clustered into four groups. San Antonio and Austin stand on their own in the centre of the state, and then Fort Worth, Dallas, Irving, Frisco, and McKinney are grouped in the upper right, and Houston, Pearland, and Conroe are grouped in the lower right.

Figure 5.12 shows how the top ten places vary in population growth over 2000 to 2020. Houston had the largest population in 2000, 2010, and 2020. The graph shows that San Antonio and Dallas are similarly sized and grew at similar rates as each other, and then with smaller populations by about 400,000 people, Austin and Fort Worth are similarly sized and grew at faster rates than the aforementioned cities. The following analysis will dive into two the cities of Dallas and Houston to understand the dynamics of local policies, population, housing, and climate risk at the city level.

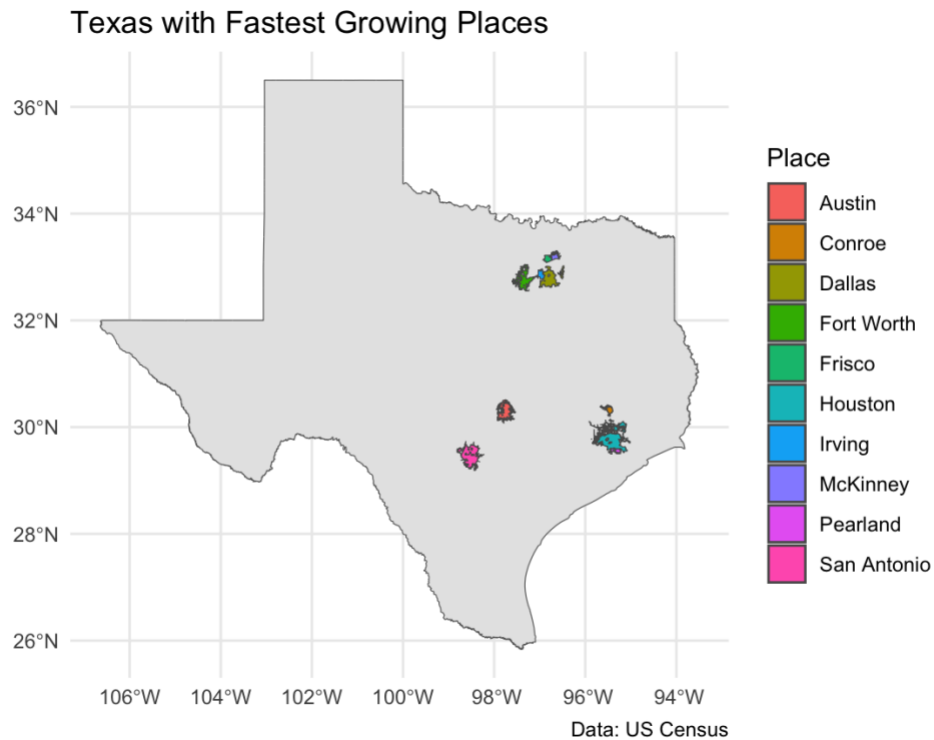


Figure 5.11: Map of Texas with the geographical boundaries of the ten fastest growing cities by absolute population change from 2010 to 2020. (Data Source: US Census)

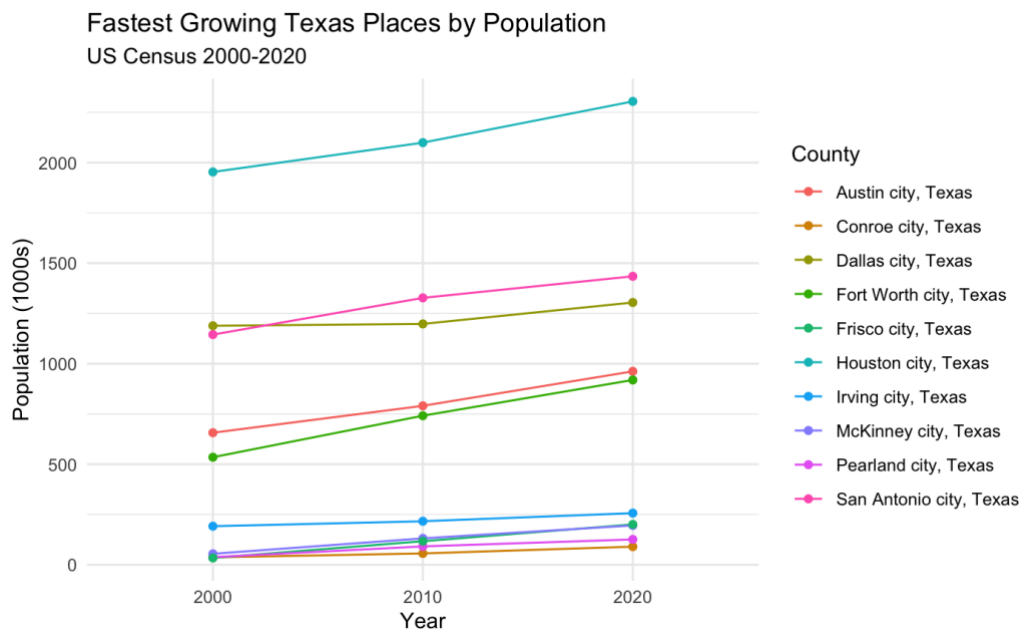
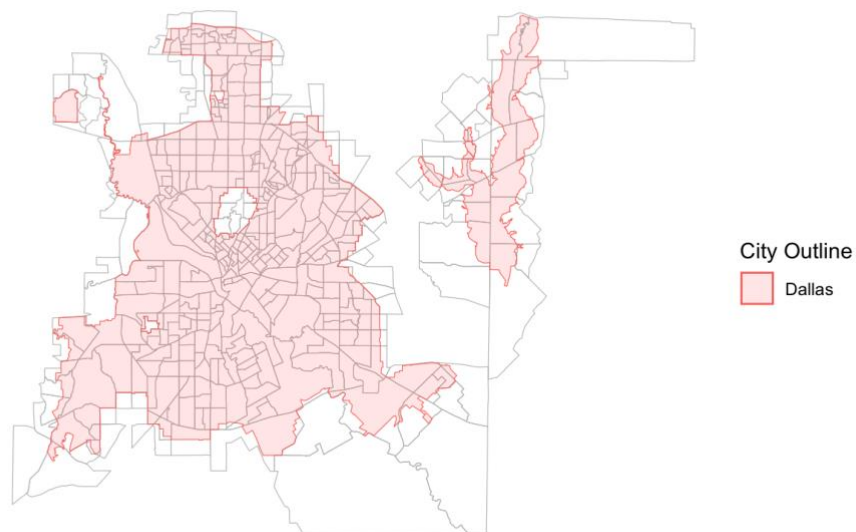


Figure 5.12: Graph of population over 2000 to 2020 for the fastest growing cities. (Data Source: US Census)

City 1: Dallas

As of 2020, Dallas is the third most populous city in Texas with an official population of 1,304,379 people (US Census). From 2010 to 2020 the City of Dallas saw an overall population increase of 8.9%, as listed in Table 5.3. Dallas is geographically close to several other of the fastest growing places shown in Figure 5.11: Fort Worth, Irving, Frisco, and McKinney. From this it is evident the larger metropolitan region has been growing considerably.

Dallas City Boundaries and Census Tracts



Data: US Census 2020

Figure 5.13: Map of Dallas showing city boundaries and census tracts. (Data Source: US Census)

Dallas has zoning policies that divide the city into areas allowing or forbidding residential development. By land area, 58.4% is residential zoning, but only 7.17% of that land allows multi-unit development, such as duplexes or flat blocks. That is 4.19% of the city land overall that allows multi-unit residential development, so the area is quite restricted.

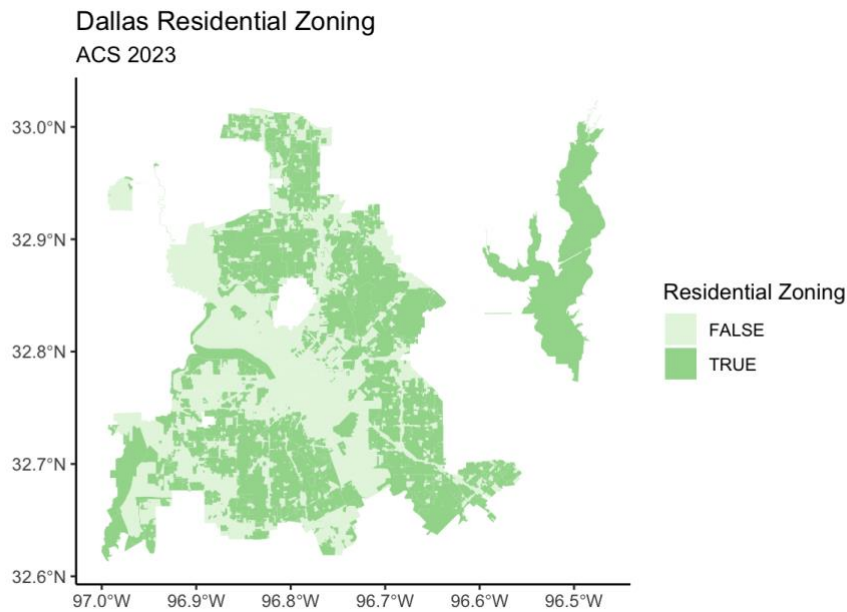


Figure 5.14: Map of residential zoning areas in Dallas. (Data Source: City of Dallas)

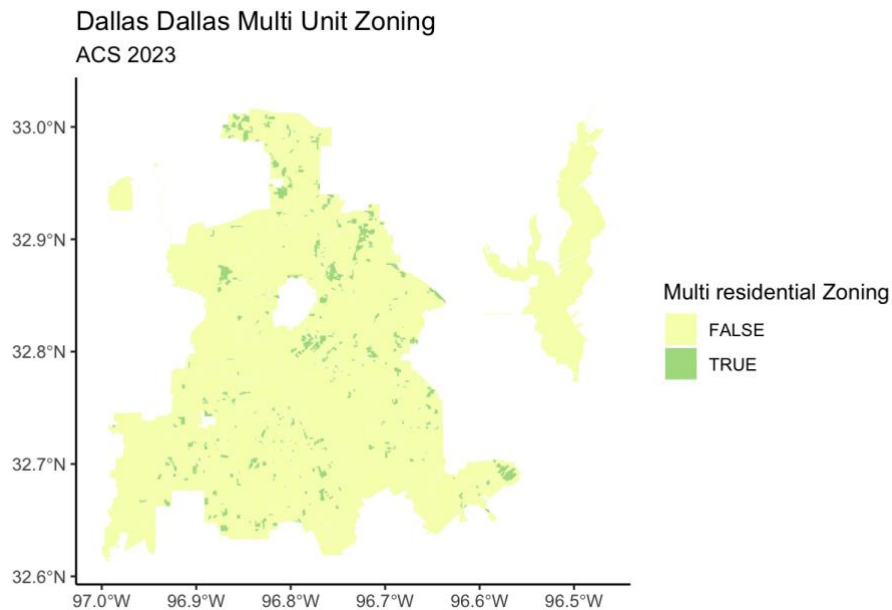


Figure 5.15: Map of multi-unit residential zoning areas in Dallas. (Data Source: City of Dallas)

As shown in Table 5.4, from 2010 to 2020, the population grew by about 9% (106,563 people) and housing units grew by about 10% (64,824 additional units) in Dallas. The housing growth percentage is just barely higher than the increase of population by percentage. These additional units were concentrated in the west and centre of the city as seen in Figure 5.16. After this period of growth, in 2020, the city had areas of overall higher housing density per square kilometre in the north and centre as mapped in Figure 5.17.

Table 5.4. Dallas population and housing summary statistics

	2010	2020	Pct Change from 2010 to 2020
Population	1,197,816	1,304,379	8.90%
Housing	637,017	701,841	10.18%

(Data Source: US Census)

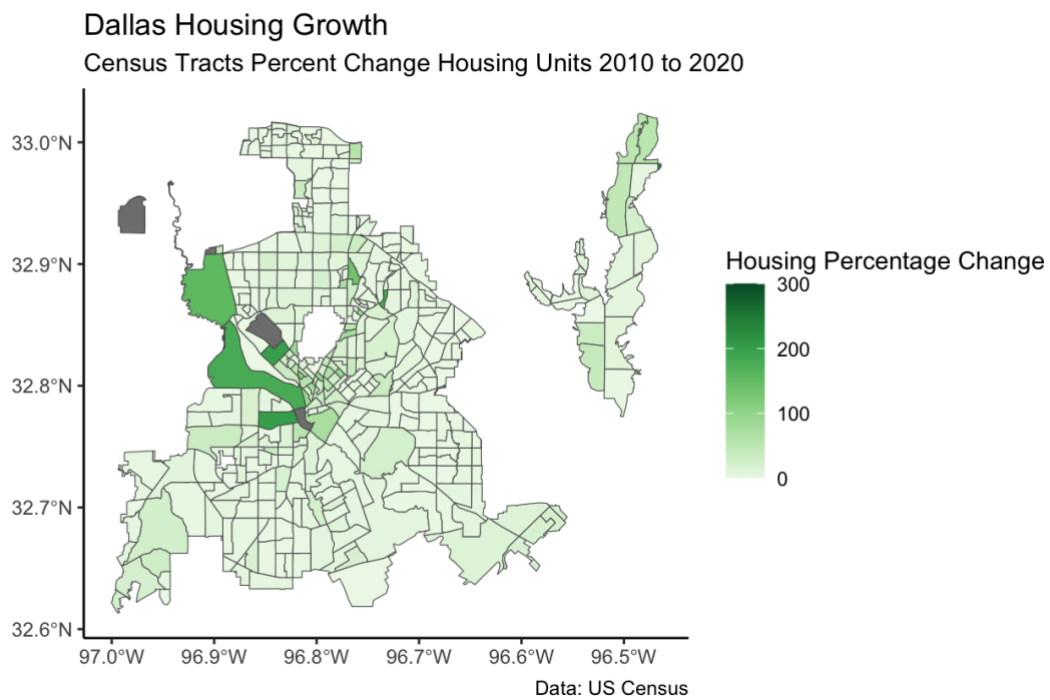


Figure 5.16: Map showing percent change in housing units in Dallas over 2010 to 2020.
(Data Source: US Census)

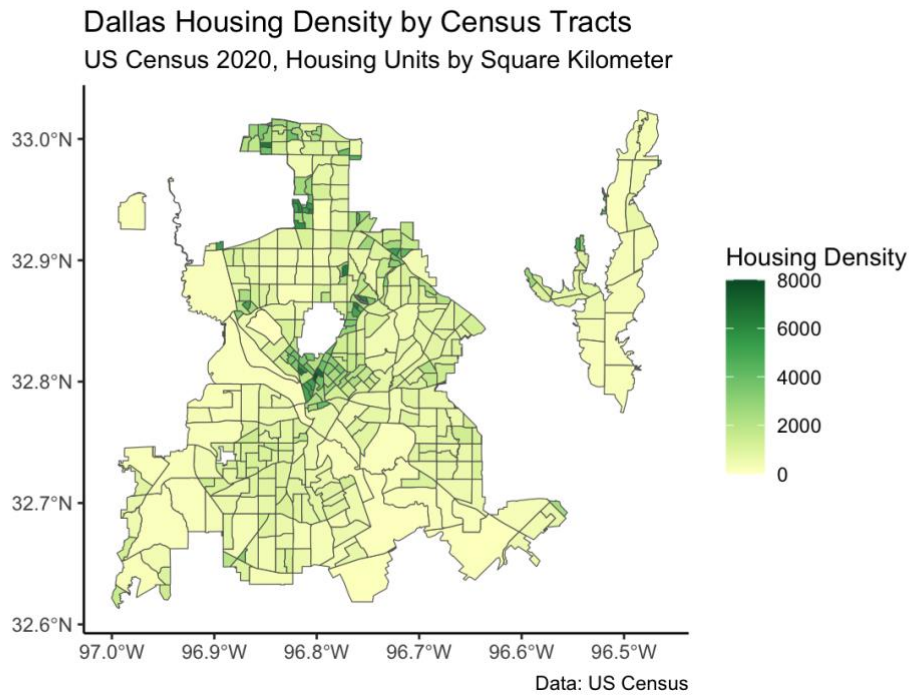


Figure 5.17: Map of Dallas housing unit density in 2020. (Data Source: US Census)

Most of Dallas's land area is categorized as either 'relatively moderate' or 'relatively low' by the NRI, as visually shown in Figure 5.18 and listed by percentage in Table 5.5. Dallas has no 'very high' NRI risk areas. As of 2020, most of the population lives in 'relatively moderate' or 'relatively low' areas. About 10.5% of the land area is in a 'relatively high' zone, which is twice as much as the percentage of population that lives in those areas according to Table 5.5. However, when looking at the distribution of building value reported by the NRI dataset in Table 5.5, 12% of the total building value of Dallas is attributed to that 'relatively high' area, which makes up 10.5% of the land area. It is notable that more than a tenth of the real estate value is in this relatively high-risk zone.

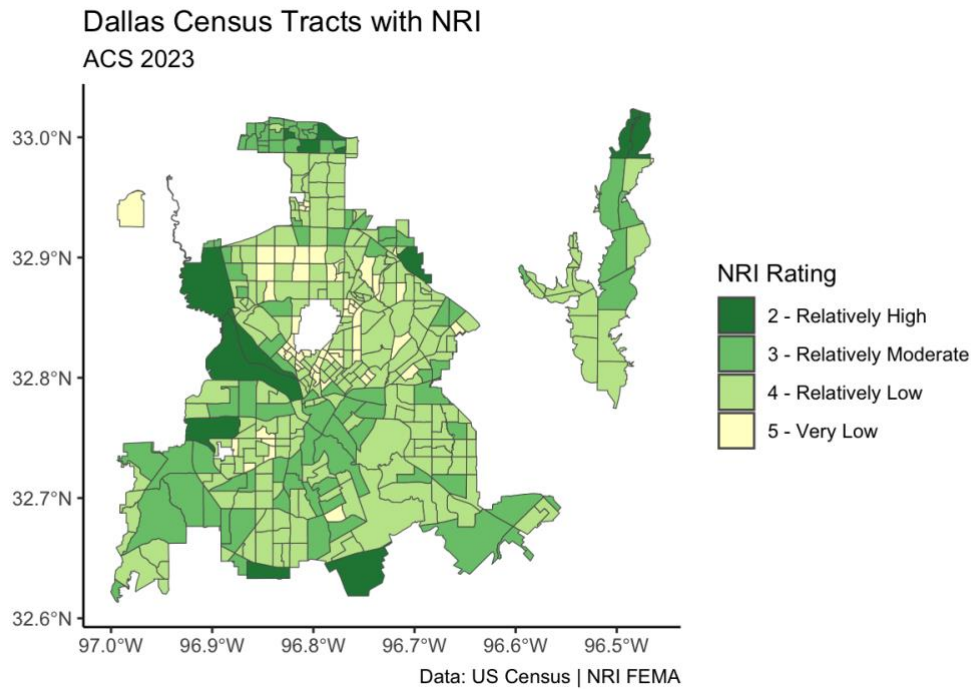


Figure 5.18: Map showing the NRI rating across Dallas by census tracts. (Data Source: US Census, FEMA)

Table 5.5. Breakdown of Dallas statistics by NRI

NRI Rating Composite	Total Area (Million sqm)	Percent Area	Population 2020	Percent Population	Building Value, \$ Billions (2023)	Percent Building Value
Very High	0	0%	0	0%	0	0%
Relatively High	105.1	10.56%	72,743	4.45%	30.9	12.26%
Relatively Moderate	310.3	31.21%	469,613	28.8%	82.1	32.60%
Relatively Low	517.0	52.00%	955,600	58.5%	119.0	47.28%
Very Low	61.9	6.23%	135,137	8.3%	19.8	7.86%

(Data Source: FEMA)

In Dallas, residential zoned land makes up a total of 584.5 square kilometres (584,466,880 square meters). By overlapping the zoning districts with the NRI data, shown in Figure 5.19, it is found that about 4 percent of residential land is in the ‘relatively high’ NRI category, as

listed in Table 5.6. The distribution of population across NRI categories listed in Table 5.5 roughly matches the distribution of Residential Zone Area across NRI categories listed in Table 5.6.

Multi-unit residential zoned land makes up a total of 41.7 square kilometres (41,682,463 square meters), and it is mapped in Figure 5.20, showing to be quite dispersed across the city. About 2 percent is in the ‘relatively high’ NRI category. Compared to the NRI risk distribution of overall land in Dallas, both the residential and multi-unit zones skew towards lower risk.

Table 5.6 Distribution of residential and multi-unit zoned land in Dallas to NRI by area

NRI	Residential Zone Area	Multi-Unit Zone Area
Very High	0%	0%
Relatively High	3.67%	2.20%
Relatively Moderate	28.71%	33.70%
Relatively Low	61.26%	58.37%
Very Low	6.36%	5.73%

(Data Source: FEMA, City of Dallas)

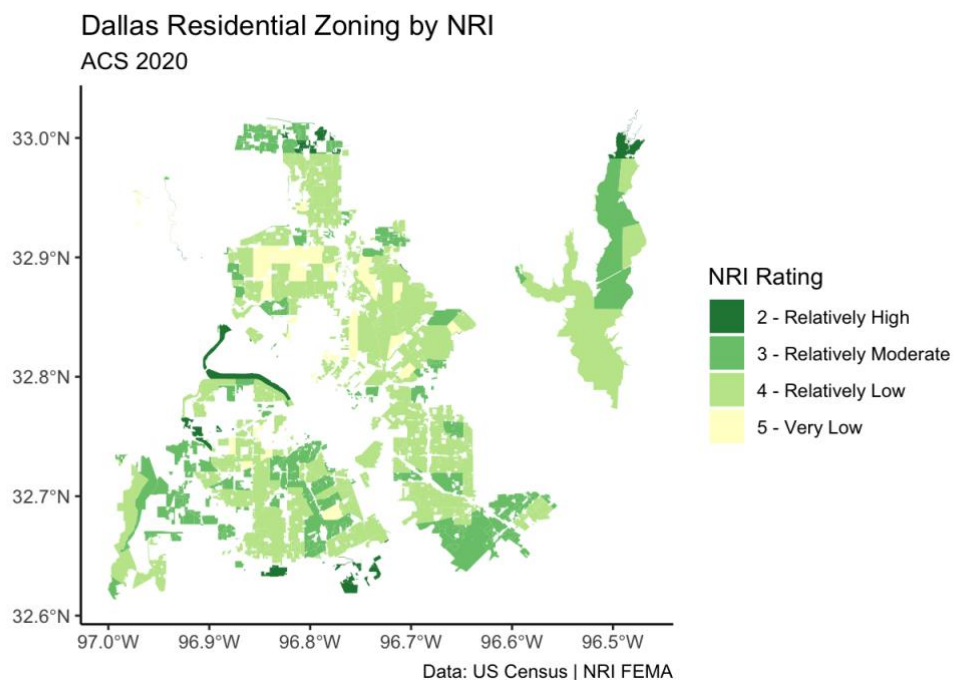


Figure 5.19: Map of area in Dallas zoned for residential development by NRI rating. (Data Source: FEMA, US Census, City of Dallas)

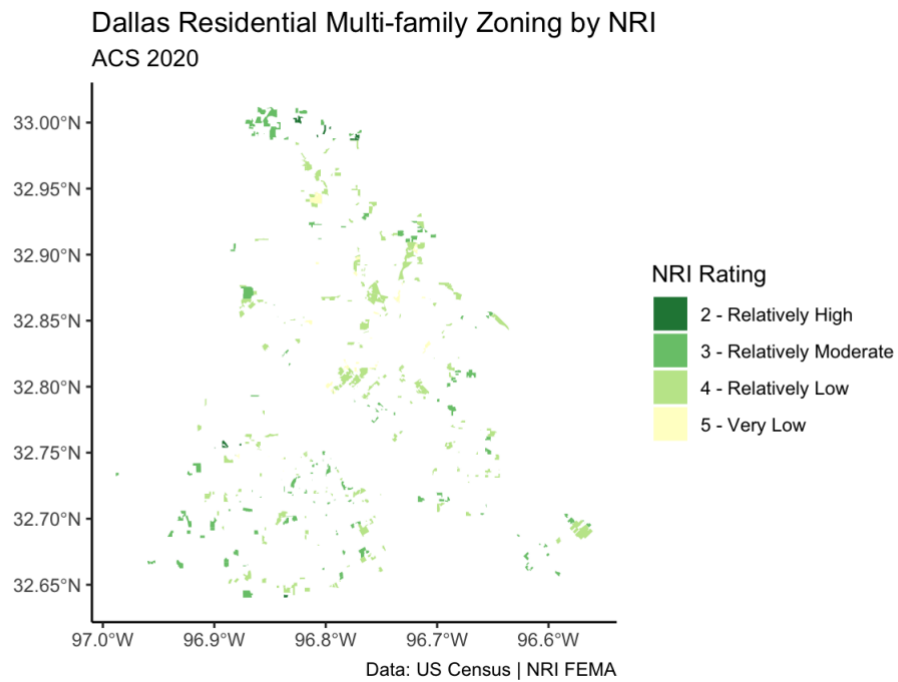


Figure 5.20: Map of area in Dallas zoned for multi-unit residential development by NRI rating. (Data Source: FEMA, US Census, City of Dallas)

Dallas had an increase of 64,824 housing units from 2010 to 2020. If the percentage of growth matched the distribution of housing in 2010, then the expected new housing growth rate in ‘relatively high’ NRI rating would be roughly 3.7% and roughly 9.7% in the ‘very low’ rating, as listed in Table 5.7. However, in Table 5.7 it shows 8.84% of new housing was built in those areas and 18.6% of new housing was built in very low areas. Therefore, it is notable that high proportions of new houses have been built in high-risk areas since 2010.

Table 5.7 Breakdown of Dallas housing by NRI

NRI Rating Composite	2010 unit count	Percent housing 2010	2020 unit count	Percent housing 2020	Unit growth 2010 to 2020 (absolute count)	Percent of total growth
Very High	0	0%	0	0%	0	0%
Relatively High	23,764	3.7%	29,495	4.2%	5,731	8.84%
Relatively Moderate	180,797	28.2%	198,077	28.2%	17,280	26.66%
Relatively Low	370,920	58.2%	400,670	57.1%	29,750	45.89%
Very Low	61,536	9.7%	73,599	10.5%	12,063	18.6%

(Data Source: FEMA, US Census)

City 2: Houston

As of 2020, Houston is the most populous city in Texas with an official population of 2,304,580 people (US Census). From 2010 to 2020 the City of Houston saw an overall population increase of about 9.77%, as listed in Table 5.8. Houston is geographically close to several other of the fastest growing places shown in Figure 5.11: Pearland and Conroe. Houston has a spread-out geography interspersed with other jurisdictions, as shown in Figure 5.21.

Houston City Boundaries and Census Tracts

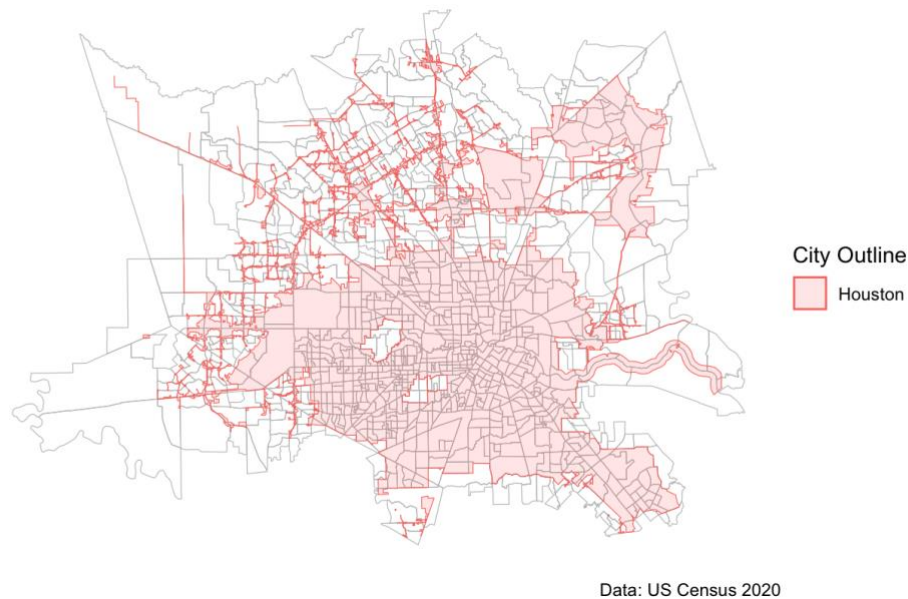


Figure 5.21: Map of Houston showing city boundaries and census tracts. (Data Source: US Census)

As discussed in the case study background in Chapter 4, Houston does not have zoning policies, so no map is included of zoning in Houston and policy restriction on residential buildings does not exist. In contrast to Dallas, single family residential development and multi-unit residential development are not restricted spatially.

As shown in Table 5.8, from 2010 to 2020, population grew by about 10% (205,129 people) and housing units grew by about 15% (227,227 additional units) in Houston. This is considerably higher than the population growth rate. These new units were spread across the city with high intensity of new units in the west, south, and northeast of the city, as seen in Figure 5.22. After this period of growth, in 2020, the city had overall higher housing density per square kilometre in the centre and west as mapped in Figure 5.23.

Table 5.8. Houston population and housing summary statistics

	2010	2020	Percent Change from 2010 to 2020
Population	2,099,451	2,304,580	9.77%
Housing	1,521,072	1,748,299	14.94%

(Data Source: US Census)

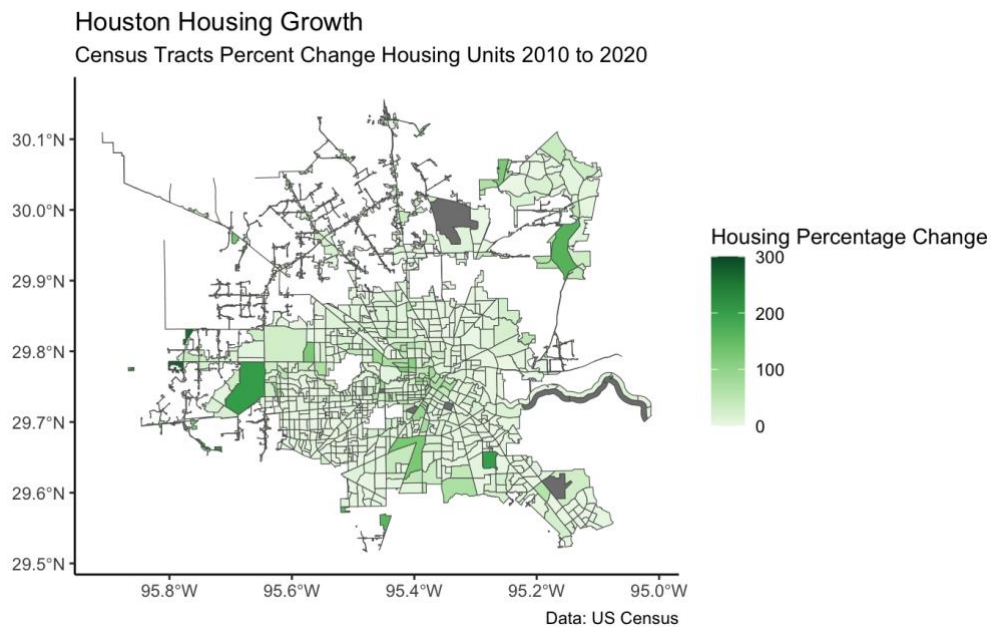


Figure 5.22: Map showing percent change in housing units in Houston from 2010 to 2020. (Data Source: US Census)

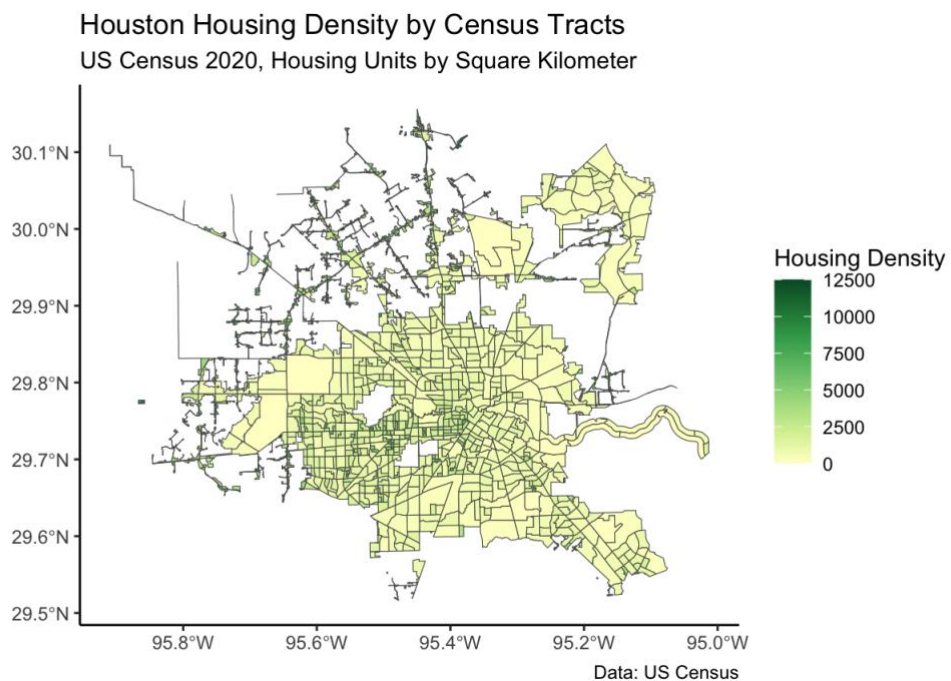


Figure 5.23: Map of Houston housing unit density in 2020. (Data Source: US Census)

The land within the Houston city boundaries is mostly considered ‘very high’, ‘relatively high’ or ‘relatively moderate’ by the NRI rating, as visually shown by Figure 5.24 and listed by percentage in Table 5.9. Houston has no land categorized as ‘very low’ NRI risk. As of 2020, most of the population lives in the ‘relatively high’ and ‘relatively moderate’ area, which together make up about 77% of the land.

When looking at the distribution of building value reported by the NRI dataset in Table 5.9, about 21% of the total building value of Houston is attributed to the ‘very high’ area, which makes up only 13% of the land area. It is notable that such a high percentage is in the highest risk zone.

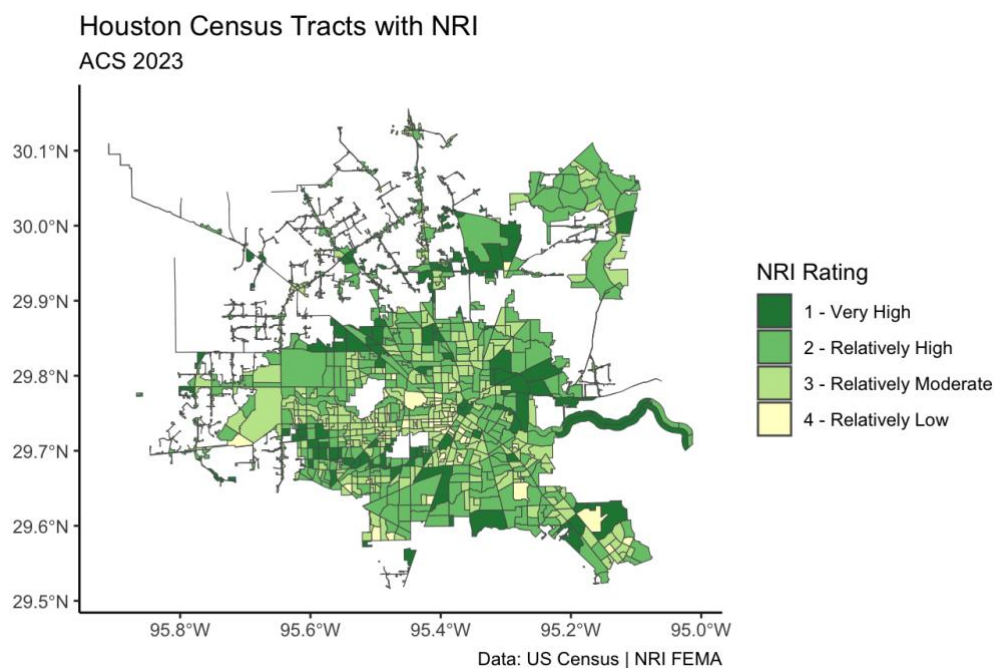


Figure 5.24: Map showing the NRI rating across Houston by census tracts. (Data Source: US Census)

Table 5.9. Breakdown of Houston statistics by NRI

NRI Rating Composite	Total Area (Million square meters)	Percent of Total Area	Population 2020	Percent of Population	Building Value, \$ Billions (2023)	Percent Building Value
Very High	329.722400	18.92%	665,583	13.35%	155.40	21.17%
Relatively High	872.895963	50.10%	2,276,360	48.87%	345.77	47.10%
Relatively Moderate	473.338682	27.17%	1,547,016	33.34%	210.51	28.68%
Relatively Low	66.492250	3.82%	159,814	3.44%	22.41	3.05%
Very Low	0	0%	0	0%	0	0%

(Data Source: FEMA)

Houston had a total growth of 227,227 housing units from 2010 to 2020, which was distributed unevenly across the five NRI rating areas according to Table 5.10. Almost 23% of the unit growth occurred in the ‘very high’ area, which only had 12% of the housing previously in 2010.

Table 5.10 Breakdown of Houston Housing by NRI

NRI Rating Composite	2010 unit count	Percent housing 2010	2020 unit count	Percent housing 2020	Unit growth 2010 to 2020 (absolute count)	Percent of total growth
Very High	179,647	11.81%	231,206	13.22%	51,559	22.7%
Relatively High	701,250	46.10%	809,777	46.32%	108,527	47.8%
Relatively Moderate	565,944	37.21%	622,882	35.63%	56,938	25.1%
Relatively Low	74,231	4.88%	84,434	4.83%	10,203	4.5%
Very Low	0	0%	0	0%	0	0

(Data Source: US Census)

Synthesis

The state of Texas, has seen back-to-back decades of rapid population growth and grown by about a third since 2000. It has seen a similar boom of housing production. The climate risk, as characterised by the NRI, is higher in the eastern part of the state, which is where several of its largest growing cities are also located, including Houston and Dallas. By analysing these two specific cities, it is apparent they both have grown quickly but have different climate risk profiles and housing growth manifested differently.

In 2010 Houston was roughly twice the size as Dallas in population. In the period from 2010 to 2020, Dallas grew by about 100,000 people, which was about 9% of their population and 10% in housing. Houston grew about 200,000 people, and 10% and 15% respectively. Over this period, Houston built about three times the housing as Dallas (227,000 versus 65,000 units).

Dallas zoning policies restrict residential development to about 60% of the land area, and multi-unit developments to 5% of the land area. Houston zoning does not exist and accordingly, housing growth was more dispersed across the city.

Houston has land with a very high NRI rating, rather than Dallas skews towards lower NRI ratings. Both Dallas and Houston have relatively large percentages of their total building value compared to percentage of population in their respective highest risk categories. In Dallas, the discrepancy was 12% versus 10.5% and in Houston, it was 21% versus 13%. This could be due to commercial non-residential buildings and therefore account for the disconnect of low relative population and high percent building value.

In Dallas, zoning policies reserve land for residential building rights that skews towards low-risk areas. Dallas saw a large percent of housing growth in its highest risk areas, with about 9% of new housing units built in the ‘relatively high’ area that previously had only 4% of the housing units in 2010. Dallas also had a large percentage of growth, about 19% in the ‘very low’ areas, which previously had 10% of the housing stock in 2010.

Similarly, Houston saw a relatively large percent of housing growth in its highest risk areas between 2010-2020. These areas rated ‘very high’ risk grew by about 23% but previously only had 12% of housing units in 2010.

Discussion

This analysis looks at how planning policies impact the risk profile of new housing units. As this research uncovers, Texas is a state with rapid house building and high climate risk; both housing production and climate risk are unevenly distributed across the state and vary within cities. By comparing Dallas, a city with stringent development policies, and Houston, a city without zoning codes, calculations found that Dallas overall has a lower risk profile of land than Houston, which has a majority of its land in ‘relatively moderate’ and ‘relatively low’ NRI areas. Additionally, Dallas’s zoning for residential land proportionally skews towards even lower risk areas. It cannot be concluded that local planning policies restrict growth to high-risk areas.

However, there are interesting findings about the risk of new housing units and overall building value of the cities, which provide takeaways for planners and communities to consider. Dallas and Houston have high rates of housing growth in the high-risk areas over the 2010-2020 period. The housing unit growth rate was approximately double the percentage of existing housing stock in those areas. Both cities have considerable proportions of total building value in their highest risk areas, which is an impactful observation when considering the risk of infrastructure investment and financial losses.

Chapter 6: Conclusion

This dissertation explored housing production and climate risk in rapidly growing cities to provide findings for planners and development stakeholders. The research utilised a case study to examine the relationship between housing development, local planning policies, and climate risk at the regional and local levels. This research adds to the literature about housing production, housing costs, and climate change as the basis of understanding the climate risk of development so that policy makers can purposefully set regional and local agendas of resilient development.

The research answered the question, ‘**Are local zoning policies restricting new housing to high climate risk areas?**’ and explored the spatial distribution of housing production and climate risk. The findings showed that there is population growth and house building clustered in several metropolitan areas in Texas. These places, including Houston and Dallas, were already main centres by 2000 and continued to expand. While it cannot be concluded that local planning policies restrict growth to high-risk areas, the research shows that new housing has been built at proportionately high rates in higher risk areas. There is also a higher proportion of total building value in the higher risk areas compared to the proportion of population in those areas. Policy makers and planners can take these findings into account to consider different strategies for building regulation in the future.

The research methodology used is repeatable for any geography within the US by using the same nationally available datasets. This offers a methodology for spatially comparing housing production and natural disaster risk and at different scales. However, there are several limitations to this research and its design. The datasets used were limited in the geographic scale they offered. The NRI was limited to counties and census tracts and did not go to a more granular scale. Additionally, the US Census changes census tracts every 10 years and redraws the boundaries such that their data makes decade over decade analysis complex. In general, the census tract is a limiting geography because they do not match up with specific risk boundaries, such as flood plains, nor city limits.

This research is also limited by scope and is not generalisable. To focus the research analysis, specific cities were chosen to illustrate contrasting planning policies and simplify spatial joining. However, a fuller analysis would consider the city and surrounding suburbs and their associated planning policies to understand the political dynamics at the metropolitan regional scale. This is important because people commute and create economic spillovers between

multiple cities. Beyond geography, the time horizon of analysis was also limited because of the scope of this dissertation to focus on the 2000-2020 period.

This research highlights further points to consider. The literature shows property market prices can be overvalued and that flood risk disclosure laws are not universally required, meaning homeowners might not be aware of the risk that threatens their property (Gourevitch et al., 2023). The public should have information on climate risk before purchasing property that is easy to access, digestible and comprehensible. This would allow individuals to evaluate the risk and possible costs of future damage. Data mapping and visualisation tools at the property and neighbourhood level can help repackaging this information for mass consumption of this information. Zillow, a large property information search engine in the US, has collaborated with First Street, a climate risk modelling company, to provide this publicly (*Zillow, 2024*).

However, given that the risk of natural disasters impacts the value of land and housing, if risk information is available and properly priced in, the market would restructure such that lower income people might only be able to afford the properties that are in high-risk areas. This is a climate justice issue that needs to be addressed by the government to prevent economically vulnerable people from also being disadvantaged by climate vulnerability.

Further Research

The research presented focuses on a narrow topic and opens the proverbial can of worms of additional research questions. The main finding of this research is that house building is occurring at fast rates in high NRI areas. Additional research could explore why development is skewing towards these areas and understand if it is market factors, such as land prices or land availability or policy factors such as environmental regulations or zoning policies that explain this development.

Further investigations could look into policy levers for local governments to take after understanding climate risk in their area. There are examples of cities addressing or limiting development in high-risk areas (*Mendonca, 2023*).

Climate migration and understanding the dynamics of people moving within or out of regions after natural disasters is also a natural extension. For example, ‘are people who have homes damaged or destroyed then choosing to leave?’ and ‘do residents have lower climate risk housing options in another part of the state or country?’.

Policy Recommendations

Existing populations, critical infrastructure, and real estate assets are already at great risk of damage from climate events. The findings of this research show rapid development in high-risk areas, and it is alarming to consider that additional people will be living in high-risk situations. National, regional, and local government policies should try to strategically consider housing development, which would be a form of climate adaptation in the face of these extreme climate events.

There is abundant research and recommendations for communities in the face of climate events. Flood risk researchers have recommended that vulnerable land can be better communicated through increased real estate disclosure laws, portraying flood risk as more than binary, and providing historic flood events and damage reports (Hino and Burke, 2021). The Joint Center for Housing Studies (2024) calls for more funding and resources to be dedicated to housing resiliency nationwide to protect from natural hazards.

Additional policies could:

- Limit housing or general development in high climate risk areas through zoning or other spatial strategies
- Require disclosure of climate risk at start of property purchase, which would help address the amount of overvaluation of properties in high-risk areas and give buyers full risk information
- Incentivize development at the regional or country scale to skew towards lower risk areas

Texas is vulnerable to frequent weather events that are induced by climate change and regularly cause great damage. Unfortunately, much of the world's population is vulnerable to these natural disasters. Understanding spatial risk distributions and how infrastructure development compares is the first step in creating better policy and smarter development investments. Hopefully with better analysis, planners, policymakers, and developers can grow communities in smart and climate safe directions.

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Figures

Unless noted, all tables and figures were generated by the author. The code for this analysis can be found in this public [Github Repository](#).

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Appendix

Table A-1. Dallas residential zoning districts

Residential District Code	District Name	Considered Multi-Unit Zoning
A(A)	Agricultural	No
R-1ac(A)	Single Family 1 Acre	No
R-1/2ac(A)	Single Family 1/2 Acre	No
R-16(A)	Single Family 16,000 Sq Ft	No
R-13(A)	Single Family 13,000 Sq Ft	No
R-10(A)	Single Family 10,000 Sq Ft	No
R-7.5(A)	Single Family 7,500 Sq Ft	No
R-5(A)	Single Family 5,000 Sq Ft	No
D	Duplex	Yes
TH-1(A)	Townhouse 1	No
TH-1(A)	Townhouse 2	No
TH-3(A)	Townhouse 3	No
CH	Clustered Housing	No
MF-1(A) and MF-1(SAH)	Multifamily	Yes
MF-2(A) and MF-2(SAH)	Multifamily	Yes
MF-3(A)	Multifamily	Yes
MF-4(A)	Multifamily	Yes
MH(A)	Mobile Home	Yes

(Data Source: City of Dallas)